POLICY REPORT

Securing Critical Minerals for India's Energy Transition

Policy Pathways and Governance Challenges

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Abstract

The transition to low-carbon energy systems is fundamentally reshaping global resource dependencies, shifting strategic focus from fossil fuels to critical minerals. This study investigates the governance of critical minerals for energy transition in India, analysing how legal, institutional, and policy frameworks respond to emerging challenges of supply security, geopolitical concentration, and sustainability.

Focusing on key technologies such as solar photovoltaics and grid-scale energy storage, the research highlights the growing material intensity of clean energy systems and the resulting vulnerabilities in mineral supply chains. Using a doctrinal legal analysis supplemented by comparative assessment and qualitative insights, the study evaluates India's regulatory architecture, particularly in light of recent reforms to the mining sector and the identification of critical minerals.

The analysis reveals structural gaps in India's governance framework, including inconsistencies in mineral classification, limited exploration and processing capabilities, and inadequate integration of environmental and social safeguards. It further demonstrates that the current policy approach remains fragmented, with insufficient coordination across sectors and stages of the mineral value chain.

The study contributes to the emerging scholarship on critical minerals by situating governance within a broader framework of energy transition, geopolitics, and justice. It argues that securing critical minerals is not merely a question of resource availability but of institutional design and strategic policy alignment. The paper concludes by outlining pathways for strengthening governance mechanisms to ensure a resilient, equitable, and sustainable energy transition.

Introduction:

Climate change is now recognized as an imminent global threat that needs urgent and sustained action from all stakeholders across the globe. The Intergovernmental Panel on Climate Change (IPCC) came out with a special report in 2018, which highlighted that at the current rate of development, global warming is likely to reach 1.5 ◦C between 2030 and 2052 (IPCC 2018). In December 2015, parties to the United Nations Framework Convention on Climate Change (UNFCCC) signed the Paris Agreement, agreeing to keep global warming well below 2 ◦C and to pursue efforts to limit it to 1.5 ◦C. The energy sector has been the largest source of greenhouse gas (GHG) emissions making energy central to any solution aimed at reducing GHG emissions. Simply defined as "a pathway toward transformation of global energy sector from fossil-based to zero-carbon by the second half of this century" (Gielen et al. 2019; IRENA 2020), energy transition is imperative to achieve climate targets and objectives. This is reflected in the NDCs communicated by countries to the UNFCCC secretariat, where energy transition is not only a recurring feature, but out of the 193 NDCs, 148 have included quantifiable renewable energy (RE) targets (IRENA 2023b). A smooth transition to RE systems is integral to the long-term strategy for carbon neutrality trajectory that has been planned by several countries in the last few years. According to an assessment in November 2021, 140 countries had already announced net-zero targets for themselves (Climate Action Tracker 2022). The net-zero pathway will rely heavily on RE sources of electricity, and minerals will be crucial for these goals.

Historically, developed countries have been responsible for most GHG emissions – nearly 80% of historical emissions, as per some estimates (CGD 2015). However, addressing this problem needs active participation from developing countries, too, which play an instrumental role in GHG emission reduction and energy transition. The demand for energy in developing countries has grown at unprecedented rates, and most of the global energy demand in the coming years will come from developing economies, especially China and India. Today, 42.6% of the total GHG emissions globally come from China, the United States, and India (WRI 2023). However, it must be pointed out that the per capita emissions present a different picture where India, the third largest emitter in absolute terms, is not even in the top ten emitters in per capita terms. The increase in population needs more resources and energy for their basic needs and socio-

economic development. Therefore, these countries have tremendous scope for using RE to reduce GHG emissions as well as meet their ever-growing demand for energy to fuel their industrial and urbanisation growth. Many developing countries in Africa and Asia, including India, are also blessed with high RE potential for resources such as solar, wind, and hydro (Energy Data Info 2024).

After the Paris Agreement of 2015, India communicated in its NDC a goal of 175 Gigawatt (GW) of non-fossil fuel-based installed capacity by 2022. The initial goal of 175 GW was updated in 2019 when the Prime Minister announced a revised aspiration of 450 GW by 2030 at the Climate Action Summit held in New York. This goal has further been changed to 500 GW as per the announcement at Glasgow, where the Prime Minister declared that India will achieve net-zero emissions by 2070 (Government of India 2022; PMO 2021).

1.1.1. Key Technologies for Energy Transition

According to the International Renewable Energy Agency (IRENA), the aggregate REinstalled capacity of the world reached by the end of 2023 was 3,870 GW, with solar having the highest share with 1.419 GW (IRENA 2024). It must be noted that 98% of the new capacities installed in 2023 were based on wind and solar energy, 69% of which were installed in Asia (IRENA 2024). This trend is likely to continue as the International Energy Agency (IEA) forecasts that solar and wind will contribute to 96% of the RE-based installed capacity additions over the next five years as these increase by more than double by 2028 from 2022 levels (IEA 2024b).

RE sources like solar energy and wind energy are freely available and present in the country. However, renewable energy-based power generation may not occur at the same time as peak demand, raising challenges for grid stability and reliable power supply (Kacker and Srivastava 2023). Grid-based electricity storage is an essential element to overcome the issues of intermittency and variability in RE. Thus, whichever RE technology one talks of, grid-based storage capacity is needed for a smooth transition to renewables. Around sixfold increase is needed in the energy storage capacity to reach 1500 GW by 2030 globally to accommodate the planned and projected increase in RE-based installed capacity of electricity in the net-zero scenario (IEA 2024a). The energy storage market growth was the highest in 2023 as it tripled with additions of 45 GW or

97 gigawatt-hours (GWh) in one year. This record is going to be broken in 2024 and surpass 100 GWh (Nsitem 2024).

India has a total installed capacity of 180 GW of RE-based electricity (Central Electricity Authority 2023a). This includes solar, wind, hydropower, biomass and waste to energy. However, as detailed in the thesis in subsequent chapters, more than half (55%) of the total RE capacity is based on solar photovoltaics (PV), amounting to 73.3 GW of installed capacity as of December 2023 (Central Electricity Authority 2023a). The integration of RE into the grid relies on electricity storage making it an essential condition for the deployment and uptake of RE source-based installed capacity. According to Central Electricity Authority (CEA)'s Optimal Generation Mix for 2030 estimates, 60.63 GW of energy storage capacity would be needed by 2030, of which 41.65 GW would be from Battery Energy Storage System (BESS) (Central Electricity Authority 2023c).

1.1.2. Resource Needs for Energy Transition Technologies

Of all the energy transition technologies, solar PV has emerged as the most promising, especially for countries like India that have solar irradiation potential. Moreover, the recent integration trend of RE into the grid has also bolstered the importance of solar energy in the energy transition trajectory. Solar PV uses semiconducting materials and solar cells to convert the energy absorbed from sunlight into electricity. The most widely used PV technology in the world and India is the crystalline silicon PV cell, either monocrystalline or polycrystalline. Technologies such as thin-film solar PV, floating solar, tandem and bifacial solar modules are also gaining popularity due to their improved efficiency and feasibility. Cadmium telluride thin-film PV may also emerge as an alternative as US companies such as First Solar Inc. are entering into contracts and partnerships with companies providing RE solutions in India (ET Energyworld 2024).

India is likely to be one of the leading markets for grid-scale storage, mostly in the form of BESS, with the storage targets set in the National Electricity Plan and the push for renewable-based installations to meet the 500 GW target of non-fossil-based electricity (Central Electricity Authority 2023b). The most popular storage technology is lithiumion cell batteries, but in the coming decade, vanadium flow batteries will become

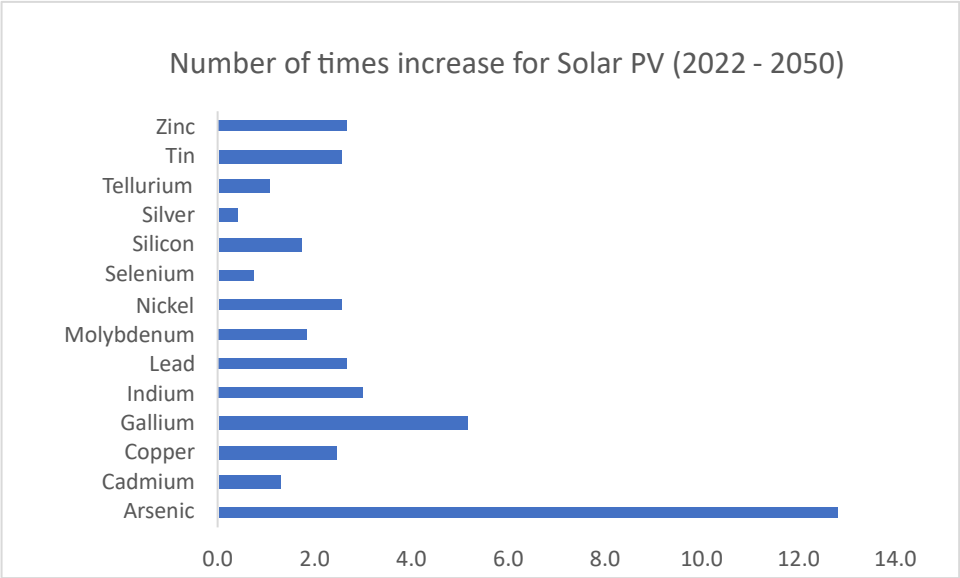
techno-economically feasible. Research on other chemical compositions is also underway, but lithium ion is likely to dominate in the coming decades.

Although meeting RE targets does not require fossil fuel minerals like coal and oil, they are resource intensive in other ways. RE-based electricity taps the energy potential of sun, wind or water but depends on other raw materials to do so, making these energy generation and storage activities extremely mineral intensive.

Table 1.1: Minerals needed for select energy transition technologies

Technology	**Key components**	**Critical minerals and material**
Solar PV	Crystalline silicon modules	Aluminium, chromium, fluorspar, magnesium, manganese, silicon, tellurium, tin, zinc
	Inverters	Aluminium, arsenic, barite, fluorspar, manganese, nickel, tin, titanium, zinc, zirconium
	Transformers	Manganese, nickel, zinc
Electrical storage	Lithium cell	Cobalt, graphite, iron, lithium, manganese, nickel, phosphorus
	Current collector	Aluminium, copper
	Vanadium flow	Vanadium
	Bipolar plates	Graphite, titanium, steel

The mineral requirement for energy transition depends on the specific technology used and the level of technological advancement. The exact amount may vary, but it is possible to foresee the demand for certain minerals for these technologies to survive (see **Figure 1.1**). Yet, it is clear that aluminium, lead, chromium, manganese, cobalt, molybdenum, copper, iron, zinc, borates, gallium, germanium, tellurium, indium, silver and silicon are needed for solar power; aluminium, indium, silicon, cadmium, iron, silver, copper, lead, tellurium, gallium, nickel, tin, germanium, rare earths, selenium, zinc are needed for wind energy; and aluminium, lithium, silicon, graphite, nickel, lead, cobalt, copper, manganese, vanadium are needed for electrical storage or batteries (Church and Crawford 2018; European Commission 2020a; IEA 2021;



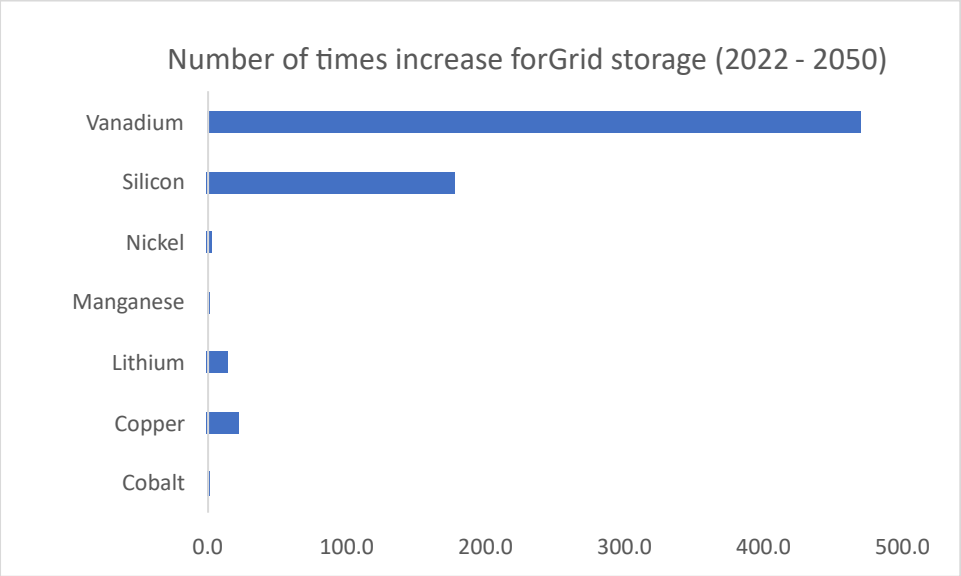


Figure 1.1: Increase in mineral demand for solar PV and electrical storage

Source: based on IEA projections

Despite India's high solar energy targets and ambition to be a manufacturing hub for solar energy equipment and domestic self-reliance (Government of India 2023), there are challenges on account of the availability of raw materials. NITI Aayog, which serves as a think tank of the government to provide strategic and technical advice on policy matters, estimates that India's lithium-ion battery industry would need 193,000 tonnes of cathode active material per annum to produce 100 GW hour of batteries by 2030

(NITI Aayog 2023a). Most of the minerals mentioned above are not available, as are processing, capacity to manufacture equipment and ancillary materials.

Energy transition

The term 'energy transition' is more than a switch in energy source and can refer to any complex, long-term, multidimensional transformation of energy systems. C*ritical minerals for energy transition (CMET)*: CMET, for the purposes of this study, refers to minerals that are needed for energy transition, as defined above. Thus, critical minerals are vital for energy technologies such as RE power technologies, EVs, energy efficiency devices, storage devices and transmission and distribution networks. There is no single definition for critical minerals, but its components are widely agreed upon – economic importance and supply security.

Solar PV: Solar PV is a modular technology that converts sunlight into electrical energy. It can be used for different applications, including rooftop PV and gridconnected utility-scale power installations. This study focuses on solar PV used in utility-scale solar-based power generation.

Electrical storage: Storage is needed for multiple energy transition technologies, including EVs. However, this study concerns grid-scale electrical storage technology. The IEA definition of grid-scale technology, that is, technologies connected to the power grid that can store energy and then supply it back to the grid, is suitable in this context.

Mining activities: The emphasis of this thesis is on the two primary activities in the upstream mineral development process – exploration and extraction. The thesis does refer to and discuss other important activities, such as processing, beneficiation and recycling but does not make it the main focus. Further, the scope of this research is restricted to underground mines and minerals in domestic jurisdictions only and does not deal with deep sea mining or international seabed mining.

Governance: Governance can refer to "a government's ability to make and enforce rules" (Fukuyama 2013:350). A wider definition of governance, as defined by the UN, guides this research. It consists of the various "mechanisms, processes and institutions,

through which citizens and groups articulate their interests, exercise their legal rights, meet their obligations and mediate their differences" (UN Secretariat 2006:3).

1.3. Relevance and Justification of the Research

Issues of supply security and the likelihood of disruptions characterise the minerals identified and listed in the chapter (Frenzel et al. 2017; Kalantzakos 2020). Many of the critical minerals needed for energy transition are not abundant and not widely or equitably distributed across the globe, making it a potential issue for international relations and geopolitics. The high concentration of these critical minerals in fewer parts of the world has resulted in high import dependence for several countries. Various studies have shown that there is likely to be a deficit of critical minerals in various scenarios (McLellan et al. 2016; The World Bank 2020; US Bureau of Energy Resources 2019).

The importance of minerals in RE technologies began to be recognised when countries started transitioning towards sustainable energy systems. The Department of Energy of the United States of America published a Critical Materials Strategy in 2011 (US Department of Energy 2011); the European Commission released a list of 14 critical raw materials as a priority action in the raw materials initiative (European Commission 2011). The Ministry of Mines, Government of India, also published a report on energycritical elements and rare earths to ensure the long-term security of national raw materials (Ministry of Mines Government of India 2012). Over the last few years, the critical minerals and energy transition interface has been prominent in domestic policies and strategies and international developments. Chapters 2 to 5 map these developments and challenges in detail and provide commentary.

1.1.3. Supply Challenges

There are concerns around the demand for critical minerals in various scenarios (McLellan et al., 2016; The World Bank, 2020; US Bureau of Energy Resources, 2019). The estimate by IEA that climate targets would entail four to six times more mineral inputs by 2040 (IEA 2021) has been called an understatement by some (DeWit 2020).

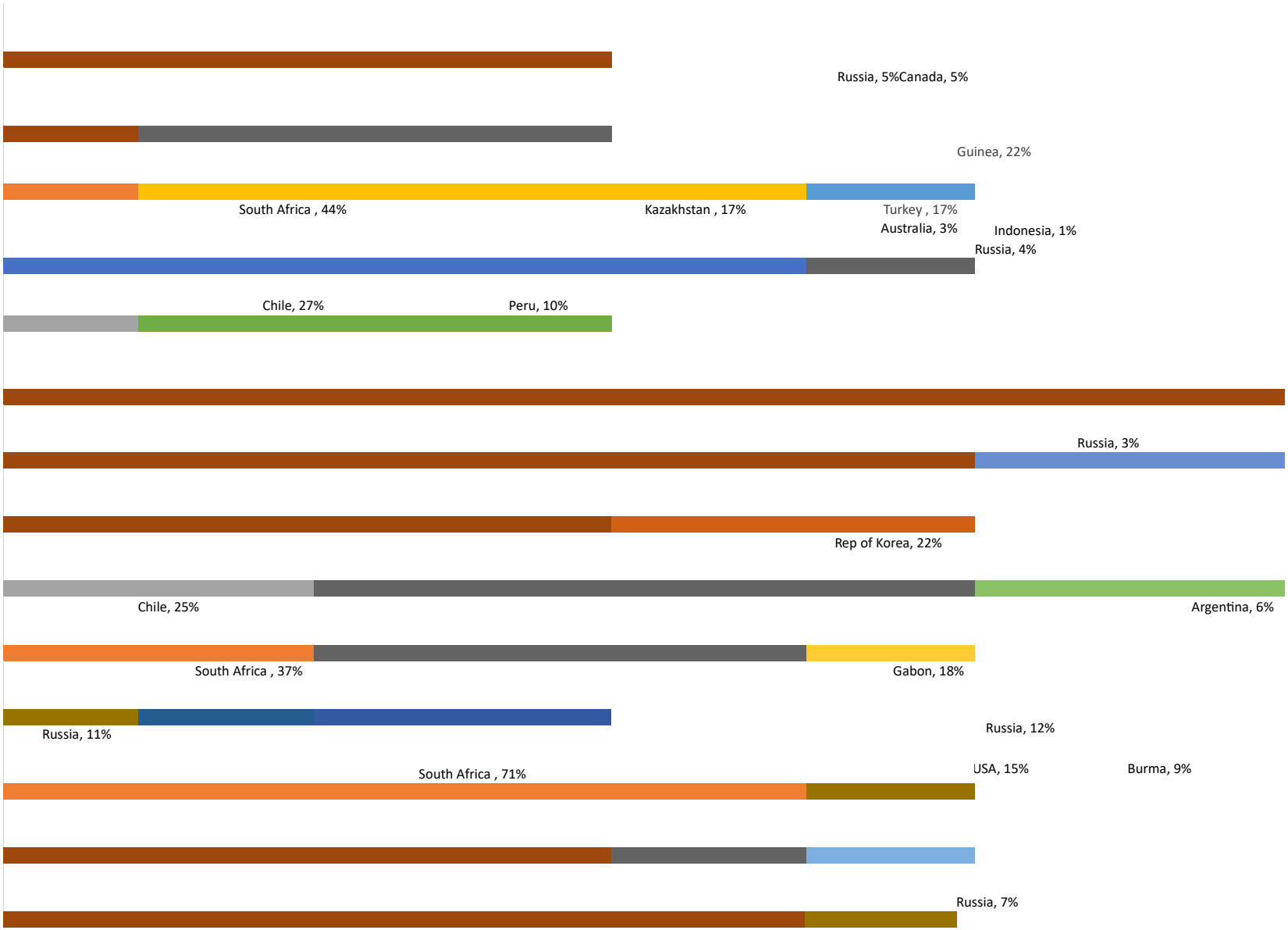
The deficit or gap in the supply chain of critical minerals needed for energy transition is due to the following:

- A mismatch between the life cycle of an energy technology and a critical mineral mine. While energy transition technologies and as mentioned in the preceding section, the deployment of RE technologies have grown at record speed, the raw materials and minerals needed cannot break records in terms of speed of extraction, especially due to uncertainties (Clare and Crawford 2018; Rietveld et al. 2019). Different mines have different lead times or start-up phases from discovery to production, but the average lead time of mines between 2002 and 2023 was 15 years (Manalo 2023).
- The natural and geographical distribution of mineral reserves is inequitable and different from the traditional sources of minerals and metals that countries have relied upon (▪ Figure **1.2**).
- The production of minerals is further concentrated in fewer jurisdictions (**Figure *1.3***).
- The ownership of the mineral chain is controlled directly and indirectly by even fewer countries. For example, more than 70% of global cobalt production comes from the Democratic Republic of Congo, but over 95% of production is done by foreign companies incorporated in China, the United Kingdom or Switzerland, with the source of control often in other countries (Leruth et al. 2022). Similarly, Indonesian nickel, which meets nearly 40% of the global demand, is extracted by foreign companies and not by Indonesian companies (Leruth et al. 2022).
- With rising demand and a limited supply chain, critical minerals have the potential to be used and misused as a trade and geopolitical tool to exert dominance.

Besides physical scarcity, lack of affordability or price volatility can increase vulnerability, especially in light of a monopolised RE supply chain (IEA 2021; Li et al. 2020; Sovacool et al. 2020) Risks emanating from trade restrictive practices (UNCTAD 2023), governance challenges in host countries (NRGI 2022), geopolitical (IRENA 2023a) and environmental risks (Watari et al. 2020a; McLellan et al. 2020) have the potential to exacerbate supply disruption further.

Figure 1.2: Distribution of critical mineral reserves across countries

Source: Calculated based on data from (United States Geological Survey 2022)


Russia, 5%Canada, 5%
Guinea, 22%
South Africa , 44%
Kazakhstan , 17%
Turkey , 17%
Australia, 3%
Indonesia, 1%
Russia, 4%
Chile, 27%
Peru, 10%
Russia, 3%
Rep of Korea, 22%
Chile, 25%
Argentina, 6%
South Africa , 37%
Gabon, 18%
Russia, 11%
Russia, 12%
South Africa , 71%
USA, 15%
Burma, 9%
Russia, 7%

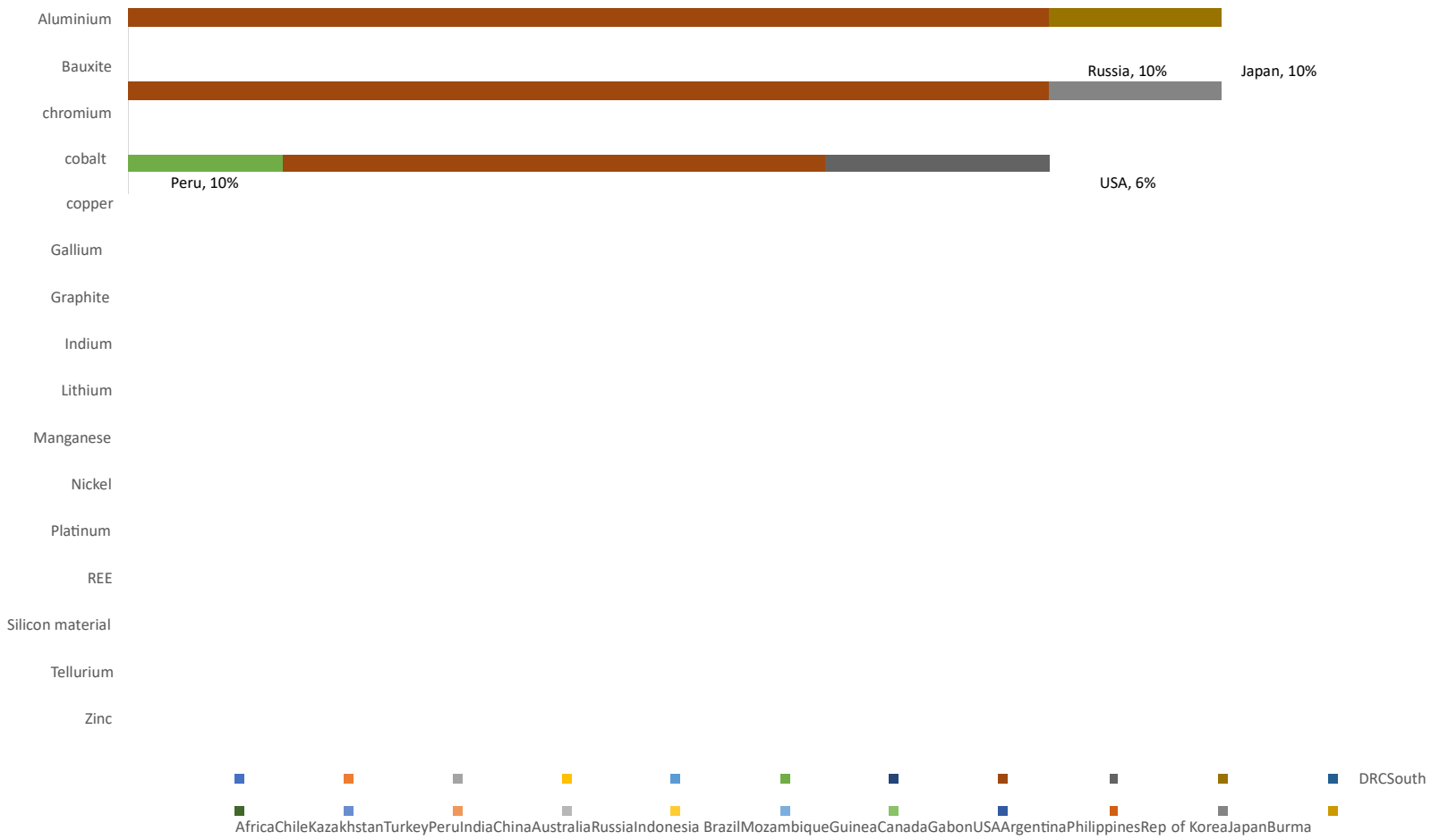


Figure 1.3: Share of countries in global production of critical minerals

Source: Calculated based on data from (United States Geological Survey 2022)

1.3.1. Geopolitics of Energy Transition

There are clear overlays of geopolitics with energy transition objectives and increased dependence on raw materials for the same. There has been an increase in literature on the geopolitical aspects of critical minerals and energy transition, especially after the Paris Agreement (Kalantzakos 2019, 2020; O'Sullivan et al. 2017; de Ridder 2013; Scholten et al. 2020; Vakulchuk et al. 2020). IRENA has also come out with a detailed report on the geopolitics of energy transition in the context of critical minerals (IRENA 2023a).

As economies move towards cleaner energy sources, RE technologies have been the bedrock of energy transition goals and plans the world over. This has been buttressed by a wider availability of RE sources in comparison with fossil fuels. However, sources of RE, such as sunlight, strong winds, tidal waves, geothermal conditions, etc. are dependent on finite resources to convert them into usable forms of electricity. These finite resources comprise metals and minerals, some of which are not widespread or easily accessible. The geographical concentration of reserves of these critical minerals needed for RE generation and storage has led to concerns of possible geopolitical tensions, as illustrated in Figure 1.4.

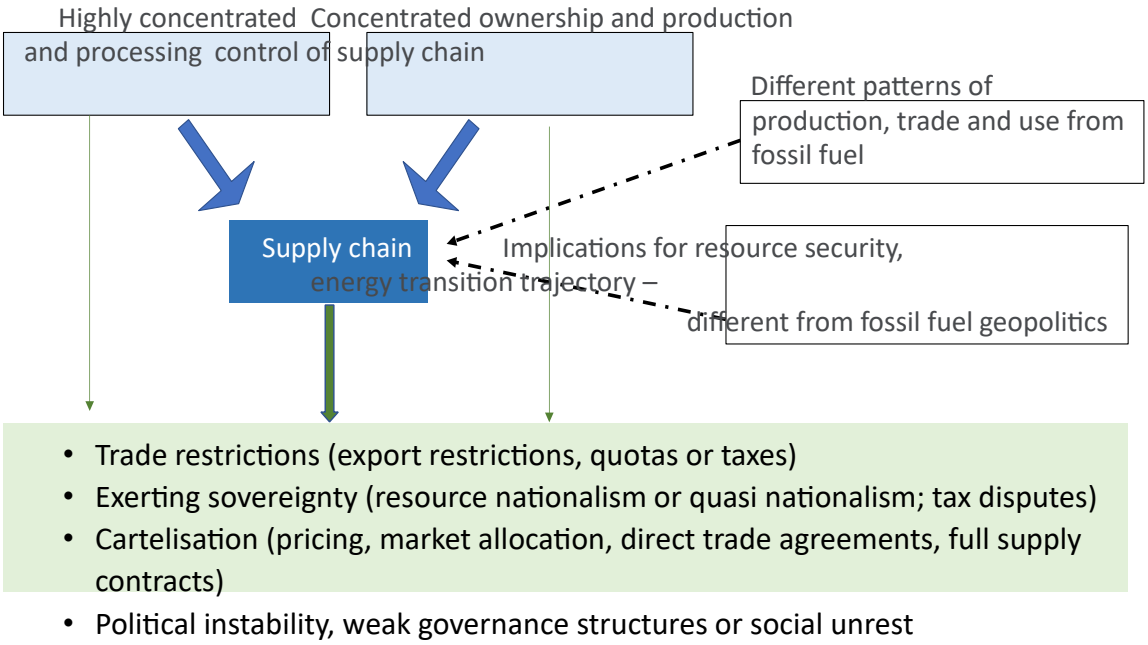


Figure 1.4: Geopolitics of critical minerals for energy transition

Geopolitical challenges in accessing critical minerals have received wider attention, especially in light of China's dominance in rare earths and other critical minerals needed for energy transition (Kalantzakos 2020; O'Sullivan et al. 2017; de Ridder 2013; Vakulchuk et al. 2020). The World Trade Organization (WTO 2015) rare earths trade dispute also underscored the possibility of disruptions and dependence of a large number of countries on China (Li, Peng, Wang, Zhang, Feng, Guan, Meng, and Wendong Wei 2020; Van Calster 2013; McLellan 2019). The role of China goes beyond trade disputes, as reflected in concerns highlighted by government actions and literature (Kalantzakos 2019, 2020). Geopolitics of renewables may be different from that of fossil fuels on account of critical materials raising newer challenges (O'Sullivan et al. 2017). In some analyses, geopolitical concerns have also been called a myth and debunked in some literature which refutes the argument that there will be increased geopolitical tension on account of critical minerals (Overland 2019).

Concentrated and near-monopolized supply chains give rise to protectionist measures, resource nationalism or high taxation, which can further exacerbate vulnerabilities and affect supply (Vikström 2020). Examples of export restrictions around critical minerals have already started emerging (Espa 2013; Espa 2015; Switzer et al. 2015). Export restrictions and trade disputes, both real and potential, have become a subject of attention in the last few years (Korinek 2019; Kowalski and Legendre 2023; Srivastava 2023; UNCTAD 2023), which led to looking at trade regimes beyond multilateral trade, such as free trade agreements and preferential trade agreements and investment treaties (Hailes 2022; Hailes and Viñuales 2024).

1.3.2. Sustainability

Any mining operation carries environmental consequences, including the creation and disposal of mine waste, alteration of land use, emission of pollutants, and contamination of water bodies (MMSD 2002). Extracting critical minerals essential for RE introduces additional environmental concerns. It leads to shifts in mining locations as different regions possess distinct mineral reserves, resulting in the emergence of new mining hubs specific to RE minerals like lithium and vanadium, which differ from those required for fossil-based energy sources such as coal and oil. Notably, lithium, vital for electricity

storage, is mainly found in select countries like Chile, Argentina, and Peru, while reserves of vanadium, crucial for vanadium redox flow batteries, are concentrated in Russia and China (IEA 2021) (The World Bank 2020). Some of these regions, like the lithium salars in South America, are ecologically sensitive, housing valuable wetlands and lagoons protected under the Ramsar Convention (Heredia et al. 2020). Mining critical minerals entails unique environmental impacts compared to traditional minerals like coal. For instance, lithium extraction is exceptionally water-intensive, leading to concerns about water depletion, especially in fragile ecosystems such as the Atacama Desert and the surrounding wetlands (Heredia et al. 2020; Lee et al. 2021). Depletion and pollution of groundwater and surface water are also issues associated with minerals like copper, nickel, silver, and rare earths (Lee et al. 2021; IEA 2021; Dominish et al. 2019). Additionally, certain atomic minerals mined in smaller quantities pose risks due to exposure to bioactive and radioactive substances (Lee et al. 2020; Wang et al. 2020).

While RE projects offer clear environmental benefits at deployment sites, the extraction of raw materials for such projects causes environmental harm elsewhere. Striking a balance between these opposing impacts is a complex task (McLellan 2020; Phadke 2020). Moreover, the lack of comprehensive data on actual impacts exacerbates the issue (Lee et al. 2020; Heredia et al. 2020; Watari et al. 2020a).

1.3.3. Justice

Some of the common social externalities on account of mining include health risks, human rights abuses, and child labour (Church and Crawford 2020). Displacement on account of mining for RE projects tends to unravel the socio-economic costs of such energy transition as well as the relationship between governance structures and the impact of the supply chain of RE projects (Kramarz et al. 2021). Concerns around the loss of habitat or change in the ownership of mines have also fuelled conflicts and confrontations between the local population and mining companies, often on issues around the lack of improved employment and other benefits for the host population (Prause 2020). It underscores how the distribution of benefits and burdens on account of energy transition is inequitable and more burdensome for countries that are already facing governance challenges. This also emphasizes the need for justice and fairness. Defined as a fair and equitable process of moving towards a post-carbon society

(Heffron and McCauley 2018), just transition is an extension of research on energy justice. McCauley et al. have defined energy justice in terms of distributional, procedural, and recognition justice (McCauley et al. 2013). The concept has been expanded to include restorative justice (Heffron and McCauley 2017).

The just transition discourse has pushed for jobs as an essential consideration in the transition away from fossil fuels and action towards skills development, social protection, active labour market policies, etc. as integrated in labour (ILO 2015) and climate proceedings (UNFCCC 2015). Although discussion on just energy transition is on one or more of the following strands – intergenerational justice, reducing vulnerability, benefits of production and consumption, compensation for damages (Hafner and Tagliapietra 2020; Hirsch et al. 2017; Pellegrini-Masini et al. 2020; Sovacool and Dworkin 2015; Zinecker et al. 2018), there is increasing recognition of the fact that mainstream definitions are not adequate to address the multidimensional issues in transition towards renewables (Bainton et al. 2021).

Discussions have revolved around minimizing negative impacts on people dependent on fossils or ensuring greater employment opportunities (Tsani 2021; Zinecker et al. 2018), but not on the impact of critical mineral extraction needed for such transition. Most critical mineral reserves are concentrated in regions characterized by prevailing governance challenges (IRP 2019), underscoring the need to address issues like the distribution of benefits and burdens ensuing from critical minerals extraction. It is imperative that raw material security for energy transition does not merely relocate environmental and social externalities from fossils to other minerals, and consequently from one jurisdiction to another. It is important to delve into why and how energy justice and just transition must be revisited in the context of critical minerals needed for energy transition.

1.3.4. Main Research Gaps and the Need for Research

In recent years, literature on mining has grown and spread beyond the traditional fields of resource course to include other areas such as environmental impact, social licence to operate (Owen et al. 2022), and critical minerals (Karakaya and Nuur 2018).

Governance of Critical Minerals for Energy Transition in India

Natural resources, especially minerals, are governed and regulated primarily by domestic governments. While international law and institutions recognise and support the ownership, sovereignty over resources and management of their various dimensions, minerals are within the domestic purview of host governments.

- Mining (process)
- Minerals (product)
- Clean Energy (end use)

The above three are distinct and have their own life cycles, but they interact with each other, as illustrated in Figure 5.1. The life cycle of a mine has exploration, development, extraction and closure. The life cycle of the product – mineral – branches out from the extraction stage of the mining life cycle. The main stages of mineral ore, after it has been extracted from under the ground, are beneficiation, processing, transport, use and recycling. The use of the mineral in RE technologies such as solar PV and electrical storage has its own life cycle but is not a subject of this chapter or thesis.

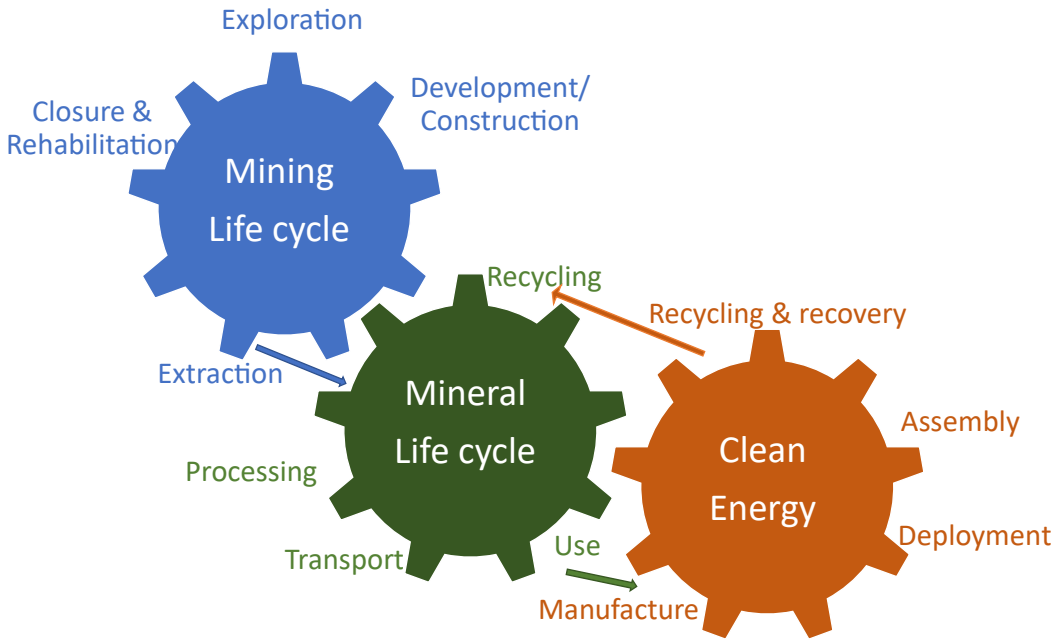


Figure 5.1: Linkages in the life cycle of mining and minerals for clean energy

This chapter presents a doctrinal analysis of national, legal and institutional frameworks for critical minerals needed for energy transition. The analysis was done for international legal frameworks and domestic regimes of other countries, before narrowing down the scope to India. Literature, primary sources like the Constitution of India, central legislation, rules, executive orders, government reports, expert committee reports, Parliament debates, data from government sources and judicial pronouncements were relied upon. Where needed, these were supported with academic literature, reports of think tanks, news reports, etc. The doctrinal analysis has been supplemented with a comparative study of the domestic frameworks of India, the US and the EU. In order to enhance validity and capture multiple dimensions of issues, findings from the document analysis and the legal analysis were triangulated with information and perspectives gained from semi-structured interviews. The interviews were kept semi-structured so as to allow flexibility.

5.2. Laws, Institutions, Policies for Critical Minerals and Mining

The legal and institutional framework drawn and explained in the previous chapter, although in the international context if applied in the domestic setting for India

would look something like Figure 5.2. The various instruments are of varying nature and degrees of influence and will be discussed in this section.

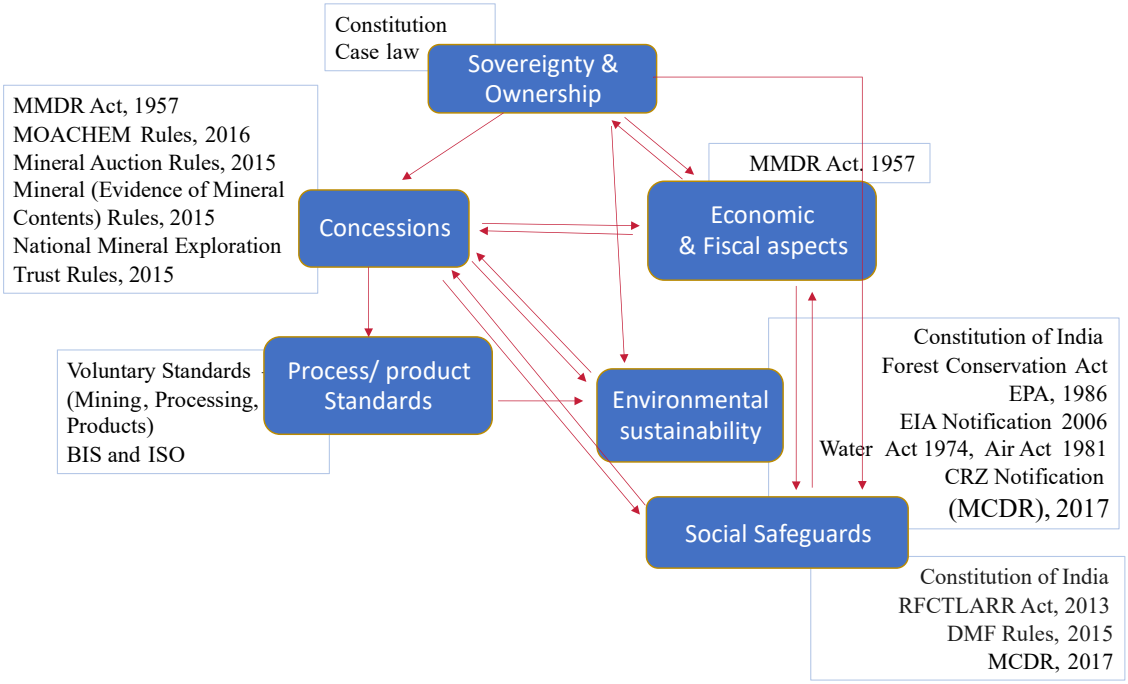


Figure 5.2: Legal framework for minerals in India

Before studying the laws, policies and institutions for critical minerals in India, it is important to first clarify what constitutes critical minerals in India.

5.2.1. Defining Critical Minerals for India

The Ministry of Mines (MoM) constituted a Committee for the identification of critical and strategic minerals headed by Joint Secretary, MoM in November 2022.[1] The committee adopted a three-stage process (Figure 5.3). Based on a review and comparison of lists of other countries, 69 minerals were identified and 29 were shortlisted at an inter-ministerial consultative meeting. The inter-ministerial meeting

included representatives from the Ministry of Power (Central Electricity Authority), the Department of Atomic Energy (DAE), IREL (India) Limited, the Ministry of

[1] vide order No. 11/1/2022-IC dated 01.11.2022

New and Renewable Energy (MNRE), the Ministry of Chemicals & Fertilizers (Department of Fertilizers) and the Ministry of Electronics & Information Technology (MEiTY). Thereafter, this list of 29 minerals was shared with the relevant ministries (the Ministry of Power, DAE, MNRE, the Department of Fertilizers, the Department of Science and Technology, the Department of Pharmaceuticals and NITI Aayog) to seek their comments.

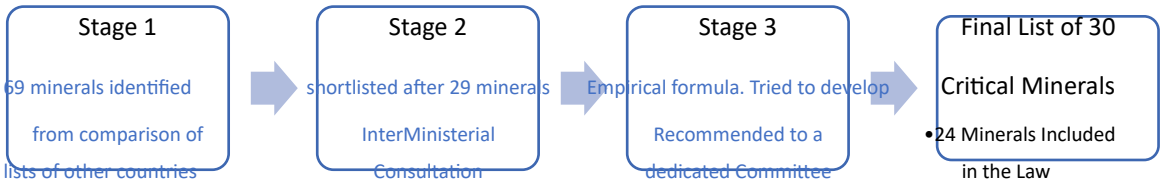


Figure 5.3: Process of identification of critical minerals in India

Source: Based on Ministry of Mines (2023b)

During the inter-ministerial consultation, the Ministry of Power listed lithium, nickel, silicon, cobalt, manganese, phosphate, natural graphite, copper, bauxite, zinc, cadmium, platinum group of elements, rare earth elements, limestone as critical for their sector. The MoM opined that India has sufficient resources of zinc, manganese, bauxite and limestone. Similarly, the MNRE listed quartz, platinum, palladium, iridium, manganese, lead, zinc, aluminium, cadmium, lanthanum, silver and yttrium. The MoM opined that India has sufficient resources of manganese, lead, zinc, bauxite and silver for now. The rest were included in the critical minerals list. Finally, a list of 30 critical minerals was finalised, comprising the following minerals (Ministry of Mines 2023b):

1. Antimony 11. Indium 21. Selenium
2. Beryllium 12. Lithium 22. Silicon
3. Bismuth 13. Molybdenum 23. Strontium
4. Cadmium 14. Niobium 24. Tantalum
5. Cobalt 15. Nickel 25. Tellurium
6. Copper 26. Tin

7. Gallium
8. Germanium
9. Graphite
10. Hafnium
16. Platinum group of elements[2]
17. Potash
18. Rare earth elements[3]
19. Rhenium
20. Phosphorous
27. Titanium
28. Tungsten
29. Vanadium
30. Zirconium

This list includes most of the critical minerals identified in the literature and databases as important for solar PV and grid-based electrical storage. According to the IEA database, the main demand for minerals in solar PV in 2022 was for cadmium, copper, indium, lead, molybdenum, nickel, selenium, silicon, silver, tellurium, tin and zinc. Similarly, the main mineral demand in grid-based storage was for silicon, copper, nickel, cobalt, lithium, manganese and zinc (IEA 2023a). The IEA envisages vanadium flow batteries to become commercially available in 2030 and grow thereafter (IEA 2023a). Out of these, lead, manganese, silver and zinc have not been listed as 'critical' by the committee as there are adequate domestic resources for these minerals (Ministry of Mines 2023b).

The Government of India made certain amendments to the main mining legislation, The Mines and Minerals Development and Regulation Act in July 2023, whereby it added a schedule of 24 'Critical and Strategic Minerals', *viz*., beryllium, cadmium, cobalt, gallium, glauconite, graphite, indium, lithium, molybdenum, nickel, niobium, phosphate (without uranium), PGE, potash, rare earths not containing uranium and thorium, rhenium, selenium, tantalum, tellurium, tin, titanium, tungsten, vanadium, and zirconium. Antimony, bismuth, copper, germanium, hafnium, silicon – minerals which were identified by the expert committee – are not included in this list included in the amended Act (See Table 5.1). This is notable for solar PV technologies, as copper and silicon have the maximum share in the mineral composition of solar photovoltaic cells. As technology evolves and substitutes become commercially usable, Gallium-Arsenide would be an effective alternative, making gallium and arsenic critical to energy transition.

[2] Platinum, Palladium, Rhodium, Ruthenium, Iridium, Osmium
[3] Lanthanum, Cerium, Praseodymium, Neodymium, Promethium, Samarium, Europium, Gadolinium, Terbium, Dysprosium, Holmium, Erbium, Thulium, Ytterbium, Lutetium, Scandium, Yttrium

Table 5.1: Global demand for solar PV and grid storage and Indian lists

	Demand (000 tonnes) (IEA 2023)				**Expert Committee on CM List**	**MMDR Act Amendment CM List**
	Solar PV		Grid Storage			
	2022	2050	2022	2050		
Arsenic		12.83			No	No
Cadmium	0.44	0.57			Yes	Yes
Cobalt			3.59	3.91	Yes	Yes
Copper	731	1811	20.4	443.8	Yes	No
Gallium		5.13			Yes	Yes
Indium	0.04	0.12			Yes	Yes
Lead	1.2	3.2			No	No
Lithium			3.44	49.68	Yes	Yes
Manganese			3.23	0.62	No	No
Molybdenum	0.15	0.28			Yes	Yes
Nickel	0.27	0.69	8.91	22.54	Yes	Yes
Selenium	0.08	0.06			Yes	Yes
Silicon	816	1402	0.07	12.4	Yes	No
Silver	4.1	1.65			No	No
Tellurium	0.46	0.5			Yes	Yes
Tin	1.3	3.36			Yes	Yes
Vanadium				470.54	Yes	Yes
Zinc	6.4	16.96			No	No

The legal and institutional framework is primarily the same as that for mines and mineral development, with some provisions and institutions specific to critical minerals.

5.2.2. Sovereignty and Ownership

Sovereignty is a broad political concept and encompasses different legal competencies, too. Simmons includes the essential manifestation of territorial sovereignty as an exercise of jurisdiction within a territory, control over land and resources, regulatory and fiscal powers pertaining to what lies within the territory, and the right to control movement (Simmons 2001). It must be noted that a state's competence on its territory or subject is two-pronged: sovereignty and jurisdiction. The distinction between *dominium* and *imperium* is valuable in appreciating the exercise of sovereignty.

In the exercise of territorial rights, ownership (*dominium*) and control (*imperium*) vested in a state and its territory must be distinguished. The ownership of minerals has attracted considerable attention from governments, companies and academia over the years (Bastidaet al. 2005). The scholarship on mineral ownership has been a reflection of the development of mining laws. Historically, it has dealt with the relationship between the ownership of land and mines and the regulation of mineral development (Wälde 1983). With an increasing need for investments in the mining sector, governments have undertaken measures to encourage and enable private investment in the minerals sector, thereby oscillating between exercising ownership and attracting private sector investments (Williams 2005). While ownership and control are important, there is a need for research on the legal regime for trade as countries depend on each other for mineral wealth. Modern international law and domestic laws locate administrative and other powers or authority of governments in *imperium* rather than *dominium* (Barral 2016; Simmons 2001). For example, in the *Thressiamma Jacob* case, in a dispute involving proprietary rights and revenue over minerals, the Supreme Court of India held that "the power to tax is a necessary incident of sovereign authority (*imperium*) but not an incident of proprietary rights (*dominium*)".[4] Therefore, sovereignty can be exercised by a state even on property that is privately owned (Viñuales 2011). Thus, in mining laws, a distinction can be

[4] *Thressiamma Jacob v. Geologist, Department of Mining & Geology and others,* (2013) 9 SCC 725, para. 51.

made between primary ownership on minerals, the right to extract resources and the extracted resource (Bastida 2020).

As discussed in the previous chapter, the newly independent resource-rich states and their people felt the need to safeguard their interests and rights over minerals and other natural resources in the new post-colonial world order and pushed for recognition of Permanent Sovereignty over Natural Resources (PSNR). Hossain writes about PSNR that, "the principle was originally articulated in response to the perception that during the colonial period inequitable and onerous arrangements, mainly in the form of 'concessions', had been imposed upon unwary and vulnerable governments" (Hossain 1984). The need to protect themselves from exploitation by foreign countries and companies culminated in demands for concerted action towards "claim to political sovereignty over the economic domain" (Pahuja, 2011:169) and right to self-

determination, seeking a framework that is an "economic corollary of state sovereignty" (Gümplová, 2014:105).

In the international negotiations for PSNR, India was not as active as other developing and formerly colonised countries of Latin America, Asia and Africa. Nevertheless, it did play an important role in suggesting amendments that were finally more agreeable to countries. In its actions closer to home, sovereignty was implied but not expressly stated in several natural resource-related decisions, such as the nationalisation of coal, oil and gas in the 1970s.

The Supreme Court of India has played an important role in applying international concepts and principles in areas of public interest for decades. While several international laws and principles have often found a way to influence court rulings, not much reliance has been placed on PSNR. There has been some passing reference in a handful of cases, but in most of the natural resource cases courts have anchored their judgements in the directive principles of the state policy. Even when upholding sovereignty on behalf of citizens, the emphasis has been on first, the distribution of natural resources to subserve the common good; second, on due process and fairness; and third, a little say on the rights of communities, especially towards

selfdetermination, an important element of PSNR. Even when the state is recognised as an owner on behalf of citizens (public trustee), the issue at stake is distribution, methods of allotment, revenue, etc. and not the right of communities to participate in the decision-making process vis-à-vis natural resources, especially energy, in which the entire nation has a stake (Srivastava 2021).

At the time of the independence, the Government of each Governor's Province before the Constitution were transferred to the Union and states, respectively. Along with the vesting of property and assets, all rights, liabilities and obligations were also transferred to the Union and state governments. With respect to offshore resources, Article 297 vests rights over all lands, minerals and other things of value underlying the ocean within the territorial waters, the continental shelf, or the exclusive economic zone with the Union. Therefore, a combined reading of Articles 294–297 of the Indian Constitution suggests that the ownership rights on all land and natural resources located within the territory of the state vest in the state and all the land and mineral resources underlying the ocean within the territorial waters, or the continental shelf, or the exclusive economic zone, inserted by the Constitution (Fortieth Amendment) Act, 1976, belongs to the Union of India. Minerals and energy resources are leased out for extraction by the government either under a concessionary regime or an auction regime. In effect, states have the right to freely exploit in any manner the natural resources within their territories (Ambrose 2013). However, the Union retains several powers and control over resources through central government laws governing natural resources such as forests, energy and non-fuel minerals (Chaturvedi 2016; Noronha et al. 2009). Governments, whether the state or Union, perform multiple functions vis-à-vis resources - owner, trustee, regulator, contractor, pursuant to the Constitution and legislation in the field of minerals, environment, finance, etc.

5.2.3. Resource Extraction and Concession

The ownership of minerals has attracted considerable attention from governments, companies and academia over the years (Bastida et al. 2005). Scholarship on mineral ownership has been a reflection of the development of mining laws. Historically, it has dealt with the relationship between ownership of land and mines

and the regulation of mineral development (Wälde 1983). With an increasing need for investments in the mining sector, governments have undertaken measures to encourage and enable private investment in the minerals sector, thereby oscillating between exercising ownership and attracting private sector investments (Williams 2005:2012). These investments are needed for the entire supply chain but are mostly sought for exploration and extraction.

According to Thomas Wälde, "a major function of any mining law is to articulate what the government's policies towards this sector of industry are" (Wälde 1988:176). This requires a symbiotic study of the laws and policies of the sector. Various policy objectives for the mining sector, such as promoting investment, privatisation or nationalisation, revenue generation, improving efficiency, etc., are manifested through mining laws and institutions. Traditionally, mining laws lay down rights and procedures for mining concessions and establish the administrative machinery that regulates and monitors mining activities (Wälde 1988). These principles, rules and regulations typically encompass rights and obligations, procedures for obtaining permission to survey, explore and extract minerals, conditions for working the mines and settling disputes, and more recently, for ensuring sustainability (Bastida 2004).

India has a rich reservoir of mineral resources producing as many as 95 metallic, nonmetallic, fuel and atomic minerals. The Government of India has called on the mining industry to increase their contribution to the country's total gross development product (GDP) to 2.5% by 2026–27 (PIB 2023). Minerals and the impacts of mining are not equitably distributed across the country. Of the country's total non-fuel minerals, 97% are produced in only seven states (Ministry of Mines 2023a), and 92% of the total coal is produced in just six states (Coal Controller's Organisation 2020). States like Odisha and Chhattisgarh are significant contributors to the production of coal and non-fuel minerals. These states' dependence on revenue accruing from minerals makes the share of mining in the gross state domestic product much higher than other states and the country as a whole (Chakrabortyet al. 2016; Noronha et al. 2009). India has a detailed set of mining related laws and rules that regulate the minerals exploration and extraction and

incidental activities. Before discussing the main legislation, there is merit in laying down the constitutional scheme with respect to minerals.

The Indian Constitution is quasi-federal in nature, dividing matters of governance into three lists: Union, State and Concurrent (Article 246). As per the Seventh Schedule of the Constitution, the development of minerals is within the legislative jurisdiction of states and the Union. List II (State List) includes 'Regulation of mines and mineral development', albeit subject to provisions of List I (Entry 23). List I (Union List) mentions, 'Regulation of mines and mineral development to the extent to which such regulation and development under the control of the Union is declared by Parliament by law to be expedient in the public interest' (Entry 54). States are the owners of minerals in their jurisdiction, but this ownership is not absolute, as it is subject to the Parliament's power to regulate the mining sector in the public interest (Vijay Kumar and Srivastava 2015).

It is important to note that, unlike certain other natural resources, such as forests and wildlife, minerals is not an item in the Concurrent List but features in both the Union and State Lists. The shared competence over minerals has always been a subject of contention rather than cooperation and is reflected in the manner in which the central government has expanded its control over states' mineral resources through the central legislation and subsequent amendments thereunder (Noronha et al. 2009).

The main legislation governing the minerals sector in India is the Mines and Minerals (Regulation and Development) (MMDR) Act, 1957 passed by Parliament for the regulation of mines and the development of minerals under the control of the Union. The Act has been amended a number of times through the years reflecting the demands and policies of different time periods. The most notable amendments were made in 2015, which changed the minerals concession system from a 'first-come-first-serve' basis to a competitive auction. Another important addition made in 2015 was the inclusion of the District Mineral Foundation (DMF) as a non-profit body to make the mining affected population a beneficiary in the revenue stream of minerals. Pursuant to the Act, all state governments had to notify

and set up DMFs. Later, the state governments were asked to incorporate the Pradhan Mantri Khanij Kshetra Kalyan Yojana (PMKKKY) into their DMF Rules.

The primary legislation governing exploration and extraction of minerals in India is the Mines and Minerals (Development and Regulation) Act of 1957, read with various rules notified under the same. Table 5.2 lists the main mining and mineral acts and rules governing the different stages of mining and what is classified as critical minerals. These are in addition to the applicability of other laws governing other aspects, such as land, forests and the environment.

Table 5.2: Mining and mineral acts and regulations

Stage	**Instrument/ process**	**Law/policy**	**Institution**
Exploration	Reconnaissance Permit Prospecting Licence Composite Licence	• MMDR Act, 1957 • Mineral (Evidence of Mineral Contents) Rules, 2015 • Mineral Auction Rules, 2015 • Minerals (Other than Atomic & Hydro Carbons Energy Minerals) Concession (MOAHCEM) Rules, 2016	• Geological Survey of India • Indian Bureau of Mines • Ministry of Mines • State Geology & Mines Department • National Mineral Exploration Trust
Stage	**Instrument/ process**	**Law/policy**	**Institution**
		• National Mineral Exploration Trust Rules, 2015 • National Mineral Policy (NMP), 2019	

Extraction/ mining	Mining Lease Composite Licence cum lease	• MMDR Act, 1957 • Mineral Auction Rules, 2015 • MOAHCEM Rules, 2016 • Atomic Minerals Concession Rules, 2016 • Mines & Minerals (Contribution to District Mineral Foundation) Rules, 2015 • NMP, 2019	• MoM • IBM • Atomic Minerals Directorate (AMD) for Exploration and Research • State Geology & Mines Department • District Mineral Foundation
Mine planning and safety	Mine Plan	• MMDR Act, 1957 • Mineral Conservation & Development Rules (MCDR), 2017 • MOAHCEM Rules, 2016 • NMP, 2019 • Metalliferous Mines Regulations, 1961	• IBM • Directorate General Mines Safety
Closure	Mine plan Progressive Closure Plan Final Closure Plan	• MMDR Act, 1957 • MOAHCEM Rules, 2016 • MCDR 2017	• IBM • MoM • State Geology & Mines Department

5.2.3.1. Critical Minerals Exploration and Extraction

Several minerals and metals needed for RE and storage fall within the purview of the MMDR Act. However, certain critical minerals, owing to their strategic importance and use in nuclear energy, were declared as atomic minerals and prescribed substances, which are governed by the Atomic Minerals Concession Rules, 2016, under the MMDR Act and the Atomic Energy Act, 1962. From time to

time, the Government of India has declared various minerals as atomic minerals and prescribed substances in light of their strategic importance and use in nuclear energy. This list included critical minerals valuable for RE and storage, such as rare earths, minerals bearing beryllium, lithium, niobium, titanium and tantalum. In 2016, beach sand minerals, which include monazite and zircon, were added to this list.[5]

Thus, extraction of all the rare earths, lithium-bearing ores and other minerals mentioned here was allowed only under a licence obtained under the Atomic Minerals Concession Rules and with approvals from the DAE. However, in 2023, the MMDR Act was amended, and the minerals listed as atomic minerals were de-listed and removed from the control of DAE and brought under the purview of the MoM.[6] Further, a new category – 'critical and strategic minerals' – has been added to the Act (Part D, Schedule I). As a result of the 2023 Amendment and the delisting of certain atomic minerals, the government has opened up the space for private sector participation in the exploration and extraction of critical minerals. Several of these minerals, owing to their designation as prescribed substances under DAE, were restricted for public sector companies prior to the said amendment.

Further, the 2023 amendment empowers the central government to grant mining leases or composite licences for the specified critical and strategic minerals through competitive bidding-based auctions. Normally, the auctions for notified minerals are carried out by the state governments. However, these critical minerals auctions will be carried out by the central government. The government justifies this on grounds that 'authorizing the Central Government to auction concession for these critical minerals

would increase the pace of auction and early production of the minerals' (PIB – Ministry of Mines 2023).

[5] Ministry of Mines Notification S.O. 2356(E), dated the 11th July, 2016

[6] Mines and Minerals (Development and Regulation) Amendment Act, 2023 (Act 16 of 2023)

In November 2023, the Central Government issued a notification inviting bids for mining leases and composite licences for 20 blocks of critical minerals such as molybdenum, graphite, potash, lithium and nickel. Since some of these minerals occur as associated minerals, the blocks contain other ores (see Figure 5.4).

Of all the blocks advertised for auction, only 25% have advanced geological information and general exploration, giving a reasonable indication of resources (G2) (see Table 5.3). This is the reason most of the advertisements are for composite licences. The newly introduced concession – exploration licence, whereby exploration and reconnaissance licences can be granted through competitive bidding, is also available for a number of critical minerals declared by the Indian government and recognised as CMET technologies, such as copper, silver, diamond, lithium, cobalt, molybdenum, lead, zinc and cadmium (Section 10BA r/w Seventh Schedule). This is in line with the National Mineral Policy that, although in passing, calls for special attention on the exploration of energy-critical minerals (Government of India 2019:4.3). The Geological Survey of India (GSI) has done the mapping of blocks and their preliminary reporting. Around 2010-11, specific missions were created in the GSI, including the one of exploration and one on R&D. The GSI is doing well with the capacity it has, but state geology departments also need to build their capacity and be able to explore blocks independently.

Table 5.3: Stages of geological study

Category	**Kind**	**Detail**
G4	Reconnaissance	Areas of enhanced mineral potential – based on geological studies, mapping, airborne and indirect methods, inference and extrapolation
G3	Prospecting	Narrowed down areas of potential. Estimates are inferred, based on the interpretation of geological, geophysical and geochemical results. Methods include (outcrop identification, geological mapping, geophysical and geochemical studies)
G2	General exploration	Initial delineation of an identified deposit (estimate of size, shape, structure and grade) giving a reasonable indication of resource continuity. Methods include surface mapping, widely spaced sampling, trenching and drilling and limited interpolation.

G1	Detailed exploration	Detailed three-dimensional delineation of a known deposit. Information in detailed exploration can help in deciding if a feasibility study should be conducted.

Source: Indian Bureau of Mines (n.d.)

5.2.4. Fiscal Regime and Industrial Policy

A fiscal regime for the mining sector usually comprises policies and rules for sharing revenue and economic value from mineral development amongst governments, industry and citizens (Energy Resource Governance Initiative 2021). These usually include a mix of production based and profit based instruments, which can have their basis in domestic legislation or in individual contracts and agreements. The Constitution of India, in its scheme of distributing subjects between the Union and states, treats taxation on mineral rights separately (entry 50, List II of Seventh Schedule, Constitution of India) and distinctly from general mining-related entries.

The main production-based instrument, royalty, is mandated under Section 9 of the MMDR Act, whereby any holder of a mining lease has to pay the royalty for 'any mineral removed or consumed'. The royalty is paid to the state government at rates prescribed by the central government (Second Schedule, MMDR Act) under three main categories: (i) on a specific tonne basis, (ii) linked to international prices, such as London Metal Exchange (LME) and (iii) on ad valorem basis, that is. at the specified percentage of the Average Sale Price (ASP). [7]

Although the production of critical minerals will take many years even if the exploration proves techno-economic feasibility, royalty rates have been fixed by the government to encourage greater participation in auctions. Rates for CMET in this study are given in Table 5.4.

Table 5.4: Royalty rates of critical minerals for energy transition

Mineral	Royalty rates
Arsenic	*

[7] Rule 39, Minerals (Other than Atomic and Hydro Carbons Energy Minerals) Concession Rules, 2016.

Cadmium	15% of ASP
Cobalt	*
Copper	4.62% of LME
Gallium	*
Indium	*
Lead	8.5% of LME
Lithium	3% of LME
Manganese	3% of ASP
Molybdenum	*
Nickel	0.12% of LME
Selenium	*
Silicon	15% of ASP (Quartz), INR 20/ tonne (sand)
Silver	5% of LME (primary); 7% of LME (By-product)
Tellurium	*
Tin	7.5% of LME
Vanadium	20% of ASP
Zinc	9.5% of LME
REE	1% ASP
Niobium	3% ASP

* All other minerals not specified in the MMDR Act royalty @ 12% of ASP

Source: Second Schedule of MMDRA Act[8]

The other fiscal instruments are in the nature of income tax, dead rent, auction premium, etc. The premium is decided through a competitive bid. Two other statutory payments – contributions to the District Mineral Foundation and National Mineral Exploration Trust are calculated on the basis of the royalty rates applicable. Hence, the contribution to state revenue as well as these funds from CMETs like lithium, rare earth, niobium and manganese, is going to be very low as their royalty rates are a mere 1% to 3%, a meagre 0.12% for royalty.

Indirect taxes such as import and export duties also impact the minerals sector indirectly. These custom duties can be on minerals, which currently range between

[8] Mines and Minerals (Development and Regulation) Act 1957 (Act 67 of 1957)

5% and 10% (Zee Business 2023), or equipment needed for the advancement of mineral exploration, extraction or processing within the country. For example, items under the HS code 8430 – other moving, grading, levelling, scrapping excavating, tamping, compacting, extracting or boring machinery, for the earth, minerals or ores attract an import duty of 7.5% of basic import duty, 28% of integrated goods and services tax and a social welfare surcharge of around 10% (Central Board of Indirect Taxes and Customs 2023). These duties and charges can clearly impact the productivity and profitability of the minerals industry in general and new and critical minerals in particular.

In recent times, there has been a renewed wave of nationalisation in the minerals sector. With the increase in demand, countries are trying to maximise revenue and exercise greater control over their own natural resources. Changes in fiscal regimes are in this direction, especially in critical minerals. For instance, Chile introduced a bill to increase mining royalties; Brazil, Zambia, Philippines are some of the countries that are taking measures to strengthen their fiscal regime to augment their minerals revenue (Nyer and

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Marchili 2021). Renegotiation of mining contracts has been tried recently in DRC and Mongolia (Mining.com 2021). Contrary to these trends in resource-rich countries and the history of the country's resource sector, India has been moving away from nationalisation or even control of public sector companies in the critical minerals sector. For example, critical minerals, that were reserved for mining by only public sector companies, owing to their strategic importance, have been opened for private sector mining companies and put up for auction.

5.2.5. Process, Standards and Specifications

Voluntary international standards governing the environment or occupational health and safety in mining operations, such as ISO 9001, 14001, and OHSAS 18001, are adopted by several Indian mining companies, both public sector companies such as NALCO and private companies such as Vedanta. It is not clear if the product-related or specific standards for minerals, especially critical minerals, by ISO, have been adopted by Indian companies.

The Bureau of Indian Standards (BIS) dominates the Indian standards and specifications space. The BIS, formerly the Indian Standards Institution, was established in 1987 as a statutory national standards body for setting standards, conformity assessment and quality assurance of products, processes and systems. One of its core functions is to establish, publish, review and promote such Indian standards.[9] Over the years, more than 500 mining-related standards have been published by the BIS for methods of test, product specification and codes of practice. Such standards exist for renewable technology devices or components for solar PV and electrical storage too. These standards are often aligned with and have equivalence to international standards for mining as well as renewable technologies, such as standards by the International Electrotechnical Commission and ISO. The Electro-Technical Department 52 of BIS has a Committee on Electrical Energy Storage Systems to look into the standardisation of grid-integrated Electrical Energy Storage Systems. While standards so far cover technical specifications (IS 17067, 17092) and performance

safety (IS 17387), industry experts feel that the entire value chain needs to be covered for better and safer performance of battery storage systems (Saurabh 2021).

5.2.6. Recycling and Recovery

As mentioned in the previous chapter, recycling and reusing minerals are highlighted as important measures for reducing dependence on imported minerals and bridging the gap between rapidly growing demand and a relatively slow increase in the supply of critical minerals. The Government of India's National Mineral Policy of 2019 recognises that although primary minerals will meet the majority of the demand, recovery of metals through recycling can increase supply, besides having environmental and economic benefits (Para 6.1). The policy also highlights the need to engage with recycling in terms of the assessment of inter-generational equity (Government of India 2019:10). These provisions are not

[9] Section 10(1)(a), Bureau of Indian Standards Acts, 2016 (Act 11 of 2016)

specific to critical minerals, but these could help in building R&D and scaling up battery recycling infrastructure.

The NITI Aayog, in its mapping of the critical minerals supply chain for lithium-ion batteries, has also highlighted that recycling mineral-bearing scrap can help in meeting some of the demand for minerals in batteries. The NITI Aayog has also recommended that government policy must enable scaling up battery recycling infrastructure with production-linked incentives, especially for critical minerals such as lithium and nickel (NITI Aayog 2023a).

The MNRE, under its Renewable Energy Research and Technology Development Programme (RE-RTD), has identified solar PV recycling as a priority area and is planning to support research on critical minerals recovery and recycling (MNRE 2023). At a regulatory level, solar PV modules and cells fall under the purview of E-Waste (Management) Rules, 2022. Under the rules, the manufacturer has to follow the rules and Central Pollution Control Board (CPCB) guidelines for storing and processing the waste generated (Rule 12). Any recycler of solar PV modules or cells has to do so in line with guidelines for recovery as laid down by the Central Pollution Control Board (Rule 9). However, the CPCB has not come out with guidelines or standard operating procedures for the same to date. Under the MMDR Act, the central government has the power to issue directions for the promotion of recycling of materials (Section 20A), but no directions have been issued yet.

5.2.7. Sustainable Development and Minerals

Mining is inherently and traditionally an activity that has a negative impact on the environment and natural resources. The finite nature of minerals and their environmental impact, when juxtaposed against the sustainable development definition (Brundtland 1987), may make the idea of sustainable mining appear to be an oxymoron (Benson and Kirsch 2010). However, mining and minerals have the potential to improve social conditions through increased revenue and livelihood opportunities, and as seen in the case of CMET, can contribute to environmental objectives, provided the externalities are minimised and managed properly. Therefore, owing to the proximity of mining activities with the economy,

environment and society, the mining industry, governments and academia have recognised the close relationship with sustainable development (Asr et al. 2019).

Since 2006, the ICMM member companies have started following the ICMM's Sustainable Development Framework (SDF) (now ICMM principles), albeit voluntarily. The sustainable development framework has also received the attention of the Government of India since the early 2000s. A high-level committee under the chairmanship of Anwarul Hoda paved the way for mining sector reforms and recommended that the SDF should be prepared for the Indian mining sector (Hoda 2006). This was incorporated in the National Mineral Policy of 2008, which required all mining to be conducted within the parameters of SDF. A draft SDF for India was prepared in 2011 to serve as a guidance document for the mining industry to improve environmental and social performance and eventually become a common benchmark for all mining operations (Ministry of Mines Government of India 2011). It was incorporated in the draft MMDR Bill of 2011. However, the bill never saw the light of the day.

The 2015 MMDR amendment empowered the central government to issue rules and directions for the implementation and evaluation of SDF.[10] Under the Mineral Conservation and Development Rules of 2017, every holder of a mining lease engaged in prospecting, mining, beneficiation or metallurgical operations has to practice

sustainable mining and obtain a minimum of three-star rating from IBM (Rule 35) even though 'sustainable mining' is not defined. The rules also make it compulsory for mine owners to adopt protective measures for mine closure, reclamation and rehabilitation (Rule 26). The IBM star rating is based on self-reporting on four key parameters – managing impacts at the mine level, mine closure and landscape restoration, addressing social impacts, community engagement and welfare programmes and reporting. The latest Mineral Policy of 2019 envisages mining as 'financially viable; socially responsible; environmentally, technically and

[10] Section 20 A, Mines and Minerals (Development and Regulation) Amendment Act, 2015 (Act 10 of 2015)

scientifically sound; with a long term view of development; uses mineral resources optimally; and ensures sustainable postclosure land uses' (Government of India 2019:6.11).

The mining industry has to obtain clearances and approvals as per the environmental regime applicable domestically. In India, the main environmental regulations that regulate the environmental management of mining, including critical minerals, are listed in Table 5.5.

Table 5.5: Environmental regulations for upstream stages of mining

Stage	**Instrument/process**	**Law/policy**
Exploration	Forest clearance	• Forest (Conservation) Act 1980
Extraction/ Mining	Environmental Approvals (EIA, Consent to Establish, Consent to Operate)	• Environment Pollution Act, 1986 • Environment Impact Assessment (EIA) Notification 2006 • Water (Prevention and Control of Pollution) Act 1974 • Air (Prevention and Control of Pollution) Act 1981 • Coastal Regulation Zone Notification

The Constitution of India states that "that the ownership and control of the material resources of the community are so distributed as best to sub serve the common good" [Article 39(b)]. In *Sanjeev Coke Mfg. Co. v. Bharat Coking Coal Ltd*, the Supreme Court held that 'material resources of the community' goes beyond natural resources owned by the public to include all natural and man-made, public and private-owned resources,[11] an approach reiterated in several subsequent rulings.[12] (*Mafatlal Industries Ltd. v. Union of India* (1997) 5 SCC 536, *State of Tamil Nadu v. Abu Kavur Bai* AIR 1984 SC 326). It clarified that the 'distribution' in the context of Article 39 (b) encompasses all forms, such as distribution between regions,

[11] *Sanjeev Coke Mfg. Co. v. Bharat Coking Coal Ltd*, AIR 1984 SC 239

[12] *Mafatlal Industries Ltd. v. Union of India* (1997) 5 SCC 536; *State of Tamil Nadu v. Abu Kavur Bai* AIR 1984 SC 326

industries, classes and nature of ownership. In upholding the nationalisation of coal mines, the court also opined that Article 39 (b) is aligned with the philosophy of socialism in envisaging the "ownership, control and distribution of national productive wealth for the benefit and use of the community" and not for selfish ends.

Mining, being a valuable natural resource, has the potential to create economic opportunities for the people. However, the interaction between mining and society has been marked by incidents of distrust, conflict, discontent, dispossession and exploitation (Bhushan and Hazra 2008; Lahiri-Dutt 2017). Most of the mineral-rich regions in India overlap with tribal areas and forests, which are home to indigenous communities. Two legislations that have recognised the rights of communities over forest lands and the right of communities to be consulted and receive fair compensation are the Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act of 2006 and the Right to Fair Compensation and Transparency in Land Acquisition, Rehabilitation and Resettlement Act, 2013.

The judiciary has tried to balance the competing economic, environmental and social rights associated with the mineral development exercise. However, in some cases, it has upheld the rights and interests of communities (Banerjee 2020). The Supreme Court of India has recognised and given legitimacy to the power of the indigenous communities to provide or withhold consent for the mining project, for example in the *Niyamagiri* case.[13]

Mineral laws and policies have also recognised the need to recognise the impact of mining on local areas and the need for sharing of benefits. The MMDR Act in 2015 introduced a DMF to be established in every mining district to work for the interest and benefit of persons, and areas affected by mining-related operations (Section 9B). The

DMF is funded through contributions calculated based on a specified percentage of royalty by companies.

[13] *Orissa Mining Corporation vs. Union of India and Ors*. (2013) 6 SCR 881

While these regulations and the need for sustainable practices are the same as for any other minerals, a few issues may make the environmental and social management of critical mineral extraction more challenging. *Firstly*, the push to explore and extract critical minerals is much higher than other traditional minerals, which could lead to expediting due process. *Secondly*, there is a lack of expertise, data and best practices for many of these critical minerals in India as they have not been traditionally mined in the country. Such expertise and data are lacking not only for technical aspects of mining but also for environmental impacts and their management. For example, most lithiummining studies and data exist for the lithium triangle in South America, which is in the nature of brines (Gielen and Lyons 2022; Vera et al. 2023), whereas lithium deposits in India are in the nature of hard rock and clay deposits. The EIA reports and processes need more baseline information to assess, mitigate and manage the impacts of mining critical minerals, which is most likely to involve advanced drilling and machinery. There needs to be greater learning from similar deposits in China or Australia (Tianming et al. 2023) and adapt them to the specific circumstances of India. The lithium deposits that have attracted the most attention in India are the ones in Jammu and Kashmir, which are close to forests and water bodies, raising concerns about impacts on fragile ecosystems (Yashwant 2023). *Finally,* ensuring sustainable practices or ESG compliance comes at a cost. Experts consulted opined that the problem lies in emphasising or determining the socio-environmental cost only at the production stage and declaring certain critical minerals as bad, e.g., cobalt. Taking this narrow and myopic approach can lead to concerns of greenflation and the consequent supply shock. Greenflation has been defined as "the upward pressures on inflation that might arise from a green transition to a state with sustainable energy production" (Olovsson and Vestin 2023). Moreover, it can also lead to the concentration of these supplies in the hands of a few actors who are not very disciplined or transparent.

5.2.8. Reducing Import Dependence

The GoI Committee on critical minerals highlighted 15 critical minerals for which India is highly import dependent. Out of the 15 minerals, there is 100% dependence on ten minerals and more than 50% dependence on three minerals (Ministry of

Mines 2023b). Thus, reducing import dependence, even though not part of the domestic legal framework, is a key strategy area and has a relationship with the domestic regime. Reducing import dependence automatically means producing more domestically.
Hence the domestic legal regime comes into play.

The Government of India has joined the Mineral Security Partnership (MSP) in 2023. MSP is a leading consortium of governments from Australia, Canada, the European Commission, Finland, France, Germany, Italy, Japan, Norway, the Republic of Korea, Sweden, the United Kingdom and the United States, to strengthen critical minerals supply across the full value chain. Quad, or the Quadrilateral Security Dialogue, is a group of four countries – Australia, India, Japan and the United States also of note. The Quad has set up a working group and announced the Statement of Principles on Clean Energy Supply Chains in the Indo-Pacific, pillared on supply chain diversification, job creation and skilling, inter-operability of standards and policies, promoting ESG and decarbonisation solutions (Quad Climate Working Group 2023).

The other approach is bilateral through agreements. The Australia–India Economic Cooperation and Trade Agreement came into force in December 2022. The Agreement will bolster partnerships for critical mineral supply produced and processed in Australia. The partnership has identified a few target projects for lithium and cobalt so far. India is also exploring similar partnerships and agreements with other mineral-rich countries. One such agreement being discussed is with Peru, a rich reservoir of copper, silver, zinc, lead, etc.

The third approach of the Indian government for reducing import-related vulnerabilities and disruptions is through overseas acquisition of mines in resource-rich locations. Khanij Bidesh India Limited (KABIL), a joint venture of three public sector mining companies – NALCO, HCL and MECL – was formed as a public sector company under the MoM in August, 2019. KABIL has signed a few MOUs with different governments and companies over the years. However, the main achievement has been its first agreement signed with the Argentinian state-owned mining and energy company in January 2024 (Kumar 2024).

5.3. Governing the CM and ET Interlinkage: A Comparative Approach

5.3.1. Definition and Identification of Critical Minerals

The first step in a Critical Minerals Strategy is to identify minerals. Different governments and agencies adopt different definitions and methodologies to ascertain what 'criticality' means to them (See Table 5.6). This section gives a list of definitions adopted by domestic governments in India, the United States of America, the EU, Australia, Canada and the UK and whether these definitions refer to the energy domain.

Table 5.6: Government definitions of critical minerals

Country	Definition of critical mineral/ material	Energy reference
India	"Critical minerals are those minerals which are essential for economic development and national security, the lack of availability of these minerals or even concentration of existence, extraction or processing of these minerals in few geographical locations may lead to supply chain vulnerability and disruption." (Ministry of Mines 2023b)	No
US	Critical Material – "(i) is essential to the economic or national security of the United States; (ii) the supply chain of which is vulnerable to disruptions (including restrictions associated with foreign political risk, abrupt demand growth, military conflict, violent unrest, anti-competitive or protectionist behaviours, and other risks throughout the supply chain); and (iii) serve an essential function in the manufacturing of a product (including energy technology-, defence-, currency-, agriculture-, consumer electronics-, and healthcare-related applications), the absence of which would have significant consequences for the economic or national security of the United States" (Public Law 116–260, section 7002(c)(4)(A)).[14]	Yes
EU	Annex I of the proposed Act sets out the list of strategic raw materials and the methodology for selecting them, and Annex II sets out the list of critical raw materials and the methodology for selecting them.	No

[14] Energy Act 2020, US

Australia	"Metals and non-metals that are considered vital for the economic well-being of the world's major and emerging	No

Country	Definition of critical mineral/ material	Energy reference
	economies, yet whose supply may be at risk due to geological scarcity, geopolitical issues, trade policy or other factors" (Geoscience Australia 2023).	
Canada	Mineral that is "essential to Canada's economic security and its supply is threatened; or required for our national transition to a low-carbon economy; or a sustainable source of highly strategic critical minerals for our partners and allies" (Natural Resources Canada 2022).	Yes
UK	"Critical minerals are not only vitally important but are also experiencing major risks to their security of supply" (Government of UK 2022)	

As clear from the various definitions, economic importance and supply risks are the two main parameters on which governments designate minerals as critical. In Europe, the list of CRM is prepared based on a detailed methodology prepared and published by the Commission comprising a combination of two parameters: (i) economic importance for the EU economy in terms of end-use applications and (ii) supply risk measured in terms of EU import reliance on the global suppliers and its import sources, with substitution and recycling identified as risk-reducing measures (European Commission 2017). The United States adopted a two-stage process – first-stage screening by using a mean of three indicators to ascertain supply risk, production growth and market dynamics and the second stage determining the critical minerals from the first-stage screening results on the basis of in-depth analysis, relying on data and inputs from various agencies (Congressional Research Service 2019; Nassar and Fortier 2021). Such indices and statistical methodology are based on extensive data which may not always be available. Hence, countries find different ways to arrive at a critical minerals list. For instance, Canada identified its critical minerals based on consultation with the provinces, territories and industry. The Government of India shortlisted critical minerals through an inter-ministerial consultative exercise (Ministry of Mines 2023b). Minerals with

sufficient domestic resources were then removed from the tentative list to arrive at the final list of critical minerals.

5.3.1.1. Critical Minerals Lists

Nearly all the countries with a critical minerals strategy or policy published a critical minerals list in 2022 and 2023. Some of these lists are new, while others have been revised and updated in light of new demands and supply chains. For example, the EU has had a list since 2011 (European Commission 2011), as a follow-up to the EU 'Raw Materials Initiative' of 2008 (European Commission 2008), and came out with its fifth list in 2023 (European Commission 2023b). The US published a list in 2018, updated in 2022 and published a revised list in 2023. A historical analysis of Japan's strategy by the Center for Strategic and International Studies reveals that a list was prepared by an Advisory Committee on the Mining Industry as early as 1984 and continued until 2014. Recently, the list has been modified to include PGM and rare-earth elements (Nakano 2021).

Based on their respective definitions and methodologies, countries have released and updated their critical mineral lists. Table 5.7 gives an overview of the various minerals classified as critical in different country strategies and policy documents. Country lists vary in size and contain 11 (South Africa) to up to 50 (USA) mineral commodities, depending on their priorities.

Table 5.7: Government-identified critical minerals in different countries

Mineral	**Australia (2023)**	**Canada (2022)**	**EU (2023)**	**India (2023)**	**S. Africa (2022)**	**UK (2023)**	**US (2023)**
Aluminium/ Bauxite	*	*	*				*
Antimony	*	*	*	*		*	*
Arsenic	*		*				*
Baryte	*		*				*
Beryllium	*		*	*			*
Bismuth		*	*	*		*	*
Boron/Borate			*				
Cadmium				*			
Caesium		*					*

chromium	*	*			*		*
Cobalt	*	*	*	*	*	*	*
Coking Coal			*				
Copper		*	*	*	*		
Mineral	**Australia (2023)**	**Canada (2022)**	**EU (2023)**	**India (2023)**	**S. Africa (2022)**	**UK (2023)**	**US (2023)**
erbium							*
Europium							*
Feldspar			*				
Fluorspar	*	*	*				*
Gallium	*	*	*	*		*	*
Germanium	*	*	*	*			*
Graphite	*	*	*	*		*	*
Hafnium	*		*	*			*
Helium		*	*				
Indium	*	*		*		*	*
Iridium							*
Lead					*		
Lithium	*	*	*	*	*	*	*
Magnesium	*	*	*			*	*
Manganese	*	*	*		*		*
Molybdenum	*	*		*			
Nickel		*	*	*	*		*
Niobium	*	*	*	*		*	*
PGM	*	*	*	*		*	*
Phosphate Rock			*	*			
Phosphorus			*				
Potash		*		*			
REE		*	*	*	*	*	*
Rhenium	*			*			
Scandium		*	*				*
Selenium	*			*			
Silicon	*		*	*		*	
Strontium	*		*	*			
Tantalum	*	*	*	*		*	*
Tellurium	*	*		*		*	*
Tin		*		*		*	*
Titanium	*	*	*	*			*

Tungsten	*	*	*	*		*	*
Uranium		*			*		
Vanadium	*	*	*	*	*	*	*
Zinc		*			*		*
Zirconium	*			*			*

Source: Compiled on the basis of government documents from Australia (Geoscience Australia 2023), Canada (Natural Resources Canada 2022), EU (European Commission 2023b), India (Ministry of Mines 2023b), South Africa (Government of South Africa 2022), UK (Government of UK 2022), USA (US DOE 2023a).

The 2023 US list identifies the ones for energy such as 'aluminium, cobalt, copper, dysprosium, steel, fluorine, gallium, iridium, lithium, magnesium, natural graphite, neodymium, nickel, platinum, praseodymium, terbium, silicon, and silicon carbide' specifically (US DOE 2023a).

5.3.2. Strategy for Augmenting Domestic Production and Processing

Building resilient domestic supply chains is seen as one of the main means of reducing import dependency and reliance on a few countries, especially China. As highlighted in the case of the EU, increased resilience needs more inward-looking measures such as the circularity of raw materials, urban waste recycling and recovering minerals and investment in primary and secondary production of minerals within the EU, albeit with environmental safeguards (Bobba et al. 2020). All three countries included in this comparative analysis have identified or proposed measures to strengthen their domestic supply chain through increased exploration, extraction, processing, substitution and recycling. Such measures focus on enabling and incentivising private sector investments, responsible/sustainable extraction and recycling.

5.3.2.1. Exploration and Production

The United States DOE's Strategy to support domestic critical mineral supply chains aims at fostering innovation and developing technologies, enabling private sector adoption and building capacity for sustainable domestic supply chains (US Department of Energy 2021). The DOE Strategy is in response to the 2020 Executive Order on supporting the Domestic Mining and Processing Industries,

which called for agencylevel plans to expand US-level critical mineral mining and processing (US Government 2020). The US also provides tax credits on the purchase of electric vehicles with batteries containing a certain percentage of critical minerals extracted or processed in the United States (U.S. Inflation Reduction Act). These domestic content requirements are a useful tool to support the domestic industry.

The European Commission came out with a Communication on Critical Raw Materials Resilience discussing challenges, critical minerals supply and suggesting measures for increasing EU resilience. The Communication proposed an action plan for critical raw materials, which, amongst other things, aimed at strengthening "the sustainable and responsible domestic sourcing and processing of raw materials in the European Union" (European Commission 2020, 6). Under the draft CRM Regulation, the Member States have to designate a national competent authority (Art 8) as a one-stop shop for facilitating the grant of mining and processing permits and accord priority to strategic projects 'in the most rapid way possible' (Article 9).

The Indian Government, as discussed in the previous section of this chapter, has amended the primary mining regulation, MMDR Act and initiated measures to incentivise private sector participation in domestic exploration and extraction by easing out restrictions in the critical minerals and auctioning blocks for critical minerals. Domestic mining is not under any single federal agency except on federal lands. In the US, an Interagency Working Group (IWG) has been constituted to reform mining permits and oversight and promote responsible mining. The IWG, in its report, recognised that the growing demand for critical minerals will result in an increase of mine planning and permitting analyses, and therefore, there is a need to enhance technical and regulatory capacity and investment (Mineral Interagency Working Group 2023). The IWG asked the Congress to collaborate with industry, communities, environmental and labour organisations "to ensure that current exploration and development is not adversely affected and that the system enhances, not hinders, future development" (Mineral Interagency Working Group 2023:98). The EU proposes devoted institutional changes to support the domestic supply chain in the form of the Critical Raw Materials Board, and the Department

of Energy and Ministry of Mines have assumed primary roles in the US and India, respectively.

The global supply chain is such that direct as well as indirect exposure is substantial, which may not be resolved through domestic mining alone. Any marginal increase in domestic production helps, which is, however, dependent on financial and administrative support from the country's governments. Mining is a time- and resourceintensive process, and details on how the exploration, extraction and processing will meet the demands need to be planned carefully and strategically. For example, US Bipartisan Infrastructure Law invests $320 million over 5 years in the USGS Mineral Resources Program's Earth Mapping Resources Initiative to advance scientific innovation and mapping of critical minerals. EU or India do not have any dedicated funds to finance or invest in increasing their contribution to the critical minerals supply chain. The draft EU Regulation talks of introducing measures to promote investments and share financial risks and proposes a finance sub-group of the proposed Critical Materials Board to provide advisory support to strategic projects obtain finance through private finance, European or international banks/financial institutions, national institutions or EU programmes (European Commission 2023b:15).

5.3.2.2. Recycling

In light of the scarcity of critical minerals and a big mismatch between demand and supply, recycling has the potential to reduce this scarcity, even if not address it substantially. The total potential of recycling to replace mining is limited as it can meet only 10% (IEA 2021) to 20% (Simas et al. 2022) of CMET until 2040– 2050. As mentioned in Chapter 2, different CMETs have different recyclability potentials ranging from 50% (iron and steel) to less than 1% (vanadium and rare earth).

The EU aims to meet 15% of its critical raw material demand through recycling. The EU Extractive Waste Directive, the Waste Framework Directive and End-of-Life Vehicles Directive can also help in recovering and recycling critical minerals from mine waste, products and equipment. Under the draft CRM Regulation, member states are obligated to prepare a National programme on circularity containing measures for waste collection, increased reuse of products and

components, increased use of secondary raw materials in manufacturing, technology advancement for recycling and materials efficiency, and skilling of workforce (Art 25). Within three years of the Act coming into force, all products would have labels to indicate the presence of permanent magnets and if they belong to neodymium-iron-boron, samarium-cobalt, aluminium-nickelcobalt, or ferrite groups [Article 27 (1)]. The Commission may introduce a delegated supplementary Regulation after 31 December 2030 laying down the minimum share of recycled critical raw materials (recovered from post-consumer waste) to be used in products incorporating permanent magnets, such as wind energy generators, cooling generators, heat and electric motors [Article 28 (3)]. One of the pillars of the US DOE Strategy is based on investing in circular-economy approaches (US DOE 2023b). The US government is supporting battery recycling. The US Office of Manufacturing and Energy Supply Chains' Battery and Critical Mineral Recycling Program supports innovation in the reuse and recycling of batteries. Three billion USD have been allocated for investment in battery materials and battery recycling facilities through the Bipartisan Infrastructure Law (US DOE 2022).

5.3.2.3. Sustainable and Responsible Mining

In Europe, existing Sustainability Certification Schemes pertaining to critical minerals and metals can apply for recognition by the Commission, provided such schemes are fair, transparent, non-discriminatory, and consider environmentally sustainable practices, socially responsible practices and business integrity (Article 29 read with Annex IV, Draft CRM Act, 2023). The Commission does not specify any action for measuring or verification of environmental footprint in the draft CRM regulation, but reserves the right and jurisdiction of the Commission to adopt a delegated act for the same in future (Article 30). In the US, the Interagency Working Group on Mining Regulations, Laws, and Permitting has issued recommendations for increasing engagement with stakeholders, affected communities and tribes. The Mining Reforms, applicable to CMET too, aim at the minerals supply chain "through environmentally and socially responsible mining and processing projects" (Mineral Interagency Working Group 2023:4).

5.3.3. Strategy for Overseas Access

Recognising issues with excessive dependence on single suppliers amid increasing use of export restrictions, domestic governments seek to strengthen rules-based open trade in raw materials and remove distortions to international trade.

Countries are trying to invest in overseas mines for CMET. Overseas investment not only helps in having some control over the supply chain in terms of the availability of minerals but also influences mining practices. The EU CRM Act mentions the potential for the deployment of Global Gateway investment projects as a factor for prioritising partnerships. This approach is useful but needs more details on how there could be synergies between raw materials security and global gateway objectives. In India, the KABIL initiative serves the purpose of investing in overseas critical mineral mines.

In light of the dependence of the EU on imported raw materials and the concentration of supply sources, it aims to diversify imports (ensuring no more than 65% of import dependence on one country) (European Commission 2023b). EU has several examples of FTAs where duties or taxes on critical raw materials are exempted, including the EUVietnam FTA,[15] and EU Korea FTA,[16] which is a leading supplier of indium. On similar lines, some of the newer FTAs have specific chapters on 'energy and raw materials' and contain provisions prohibiting the use of export monopolies, export taxes, etc. (Crochet and Zhou 2023).

The US has signed and announced dedicated Critical Minerals Agreements, such as the US-Japan Agreement, to ensure the free trade of cobalt, graphite, lithium, manganese, and nickel.[17] A similar agreement has been proposed between the US and EU to diversify the international supply chains of critical minerals required for the production of electric vehicle batteries (European Commission 2023a). India

[15] Free Trade Agreement between the European Union and the Socialist Republic of Viet Nam (EUViet Nam) (Adopted on 12 June 2020, Entry into force on 1 August 2020) OJ L 2020 186

[16] Free Trade Agreement between the European Union and Its Member States, and the Republic of Korea (EU-Republic of Korea) (Adopted on 14 May 2011, Entry into force on 1 July 2011) OJ L 2011 127

[17] Agreement Between the Government of the United States of America and the Government of Japan on Strengthening Critical Minerals Supply Chains (US-Japan), (Adopted on 28 March 2023, entered into force 28 March 2023) TIAS 2023 328

and Australia also entered into a bilateral trade agreement removing removed tariffs for several critical minerals originating in Australia.[18]

Partnerships with existing allies and new actors are needed for critical raw materials resilience. The Commission came out with a Raw Materials Initiative in 2008 with an integrated strategy to improve access to non-energy and non-agricultural raw materials. The integrated strategy included 'Raw materials diplomacy' as a central element. Raw Material Diplomacy was reiterated in the Communication on tackling Raw Material Commodity market challenges, which prescribed that the EU to "actively pursue a 'raw materials diplomacy' with a view to securing access to raw materials, in particular the critical ones, through strategic partnerships and policy dialogues" (European Commission 2011).

In addition to sections or chapters on minerals and raw materials in trade agreements, critical mineral-specific partnerships and clubs have also emerged in recent times. Thus, using strategic partnerships to improve the supply of critical minerals whose supply chain is currently concentrated in a few jurisdictions is obvious and pragmatic. The EU, the US and India are already a member of the Mineral Security Partnership. Some more examples of such partnerships include the Australia–UK Critical Minerals Joint Working Group (2020), the US–Canada Joint Action Plan on Critical Minerals Collaboration (2020), the US–Australia (2021), and the US–Japan Partnership for Resilient Supply chains for critical minerals (2022).

5.4. Main Strategies to Secure CMET by Countries

Key components that together form the domestic legal framework for the governance of critical minerals for energy transition are the Constitution, legislation and regulations, judicial precedents, legal protections, and bilateral and international agreements. The discussion in this chapter shows that country

[18] Australia-India Economic Cooperation and Trade Agreement (Australia-India FTA) (Adopted on 2 April 2022, Entry into force on 29 December 2022)

governments essentially follow a two-pronged approach to securing and governing critical minerals to fuel their energy transition journey and goals. The two-pronged approach anchored on (a) onshoring and (b) right shoring or friend-shoring is essentially aimed at strengthening the supply security of raw materials.

5.4.1. The Two-pronged Approach

Onshoring entails decisions and policies to locate or create production activities close to market demand (Backer et al. 2016). As discussed in the case of India and comparative study of other import-dependent jurisdictions, governments see onshoring or reshoring, i.e. bringing back production to home after a period of offshoring (Bolter and Robey 2020), as a strategy to have greater control over critical minerals and raw material supply chains for energy transition technologies. The other approach is friendshoring or ally-shoring, i.e. the practice of shifting supply chain networks to countries perceived to be political and economic allies (Maihold 2022). This is important and unavoidable to reduce vulnerabilities in import supply as minerals, unlike services or manufacturing, are location specific and not possible in every jurisdiction. While it has the benefit of diversification (Vivoda and Matthews 2023), it could result in an overreliance on a few nations and even GDP loss in some cases (Javorcik et al. 2022).

5.4.2. Strengthening Pillars of CMET Governance for Solar PV and Electrical Storage

The rise of solar energy-based renewable energy capacity in India is often hailed as a success story. The solar story that has yielded results in the last decade merits attention, separate from the overall renewable energy or energy transition. The evolution of solar energy in India has been presented in Chapter 2 and this chapter focuses on the supply chain of raw materials and minerals needed for solar PV and grid-scale electric storage.

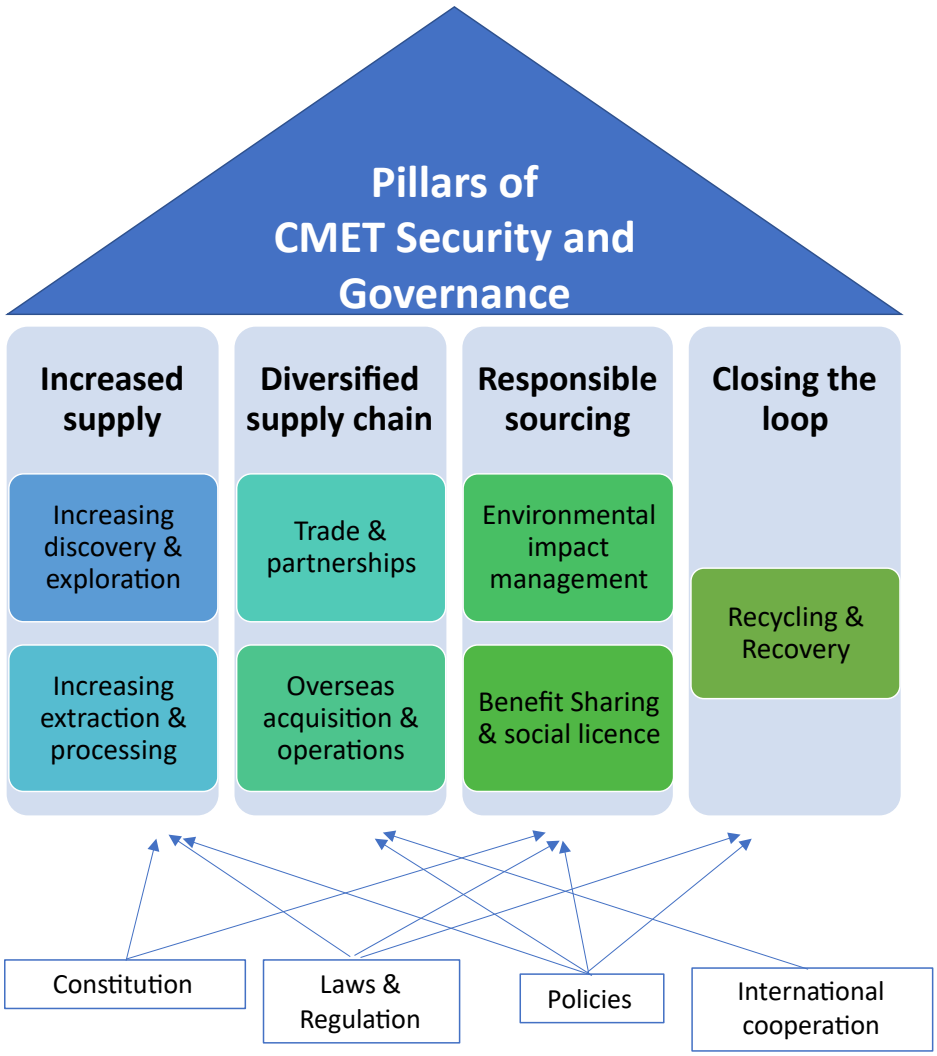

Figure 5.5: Pillars of critical minerals for energy transition governance

5.4.2.1. Increased Supply Sources

Domestically, augmenting production is one of the most important and first steps in strengthening supply sources of critical minerals needed for use in the solar PV industry and BESS.

Recognising the need for prospecting and exploration in order to determine the technoeconomic feasibility of mineral reserves and the need for incentives for the same, governments, including the Indian government, have been paying attention to studies, surveys and exploration for critical minerals. In order to attract private sector participation and investment, institutional and regulatory support is being provided in the form of preliminary surveys by government agencies like GSI, data repositories, and offering mineral blocks for exploration licence or composite

licence (exploration cum mining). However, the experience with past auctions and, in the opinion of stakeholders consulted for this thesis, the current regime and the practice of auction is counter-productive for increasing exploration and R&D or even value addition. The role of exploration was underlined by experts as most important and urgent. Many of the critical minerals for which demand will go up for solar PV and grid storage by 2050 have traditionally not been a part of GSI annual plans for exploration. However, this has been changing since the Field Year 2020–2021. Under Mission IIA on strategic and fertiliser minerals, 59 exploration projects were taken up in 2020–2021, 118 in 2021– 2022, 123 in 2022–2023 and 122 in 2023–2024 (Joshi 2023; Kumar 2023).

In the Field programme for 2023–2024, GSI has proposed mineral exploration programme for many minerals needed for the two case study energy transition technologies – tin, tungsten, molybdenum, REE (strategic minerals); bauxite, copper, lead, zinc, antimony (base metals); limestone, magnesite, graphite, pyrophyllite, etc. (industrial minerals); gold, PGE, diamond (precious metals and minerals) (GSI 2023). In response to a parliamentary question in the Rajya Sabha, the Union Minister for Mines mentioned that GSI had carried out 20 exploration projects for lithium between 2017 and 2022 in Andhra Pradesh, Arunachal Pradesh, Bihar, Chhattisgarh, Himachal Pradesh, Jharkhand, Jammu and Kashmir, Madhya Pradesh, Meghalaya and Rajasthan.
During FSP 2022–2023, a further 18 projects were initiated (Joshi 2022).

Extraction and processing of minerals are also incentivised and eased through firstly, identifying critical or strategic minerals, facilitating and streamlining the procedure for mining and processing permits, as in the EU, or having auctions and attractive and reduced royalty rates for critical minerals as in India. It must be noted that arsenic, copper, lead, manganese, silicon, silver and zinc, which are going to be needed for solar PV or electric storage, or both, are not recognised as critical minerals in the List inserted in the 2023 Amendment of the MMDR Act (See Table 5.1). Information gaps also remain as IBM, which publishes mining and minerals data, including ASP etc., for key minerals, does not collate and publish data on some of these critical minerals.

In light of the increasing interest in demand, building technical and regulatory capacity is needed across the value chain, from exploration to extraction to processing. Processing or value addition is important, if one looks at critical minerals to support energy transition technologies, and not just a source of revenue from the minerals sector. Mineral discovery and mineral processing are interrelated and whether backward linkage should be done first or mineral discovery first is difficult to answer, and, ideally the two should be developed parallelly to reduce vulnerability.

The lack of capacity to process and manufacture can continue to make the supply chain vulnerable and dependent on imports, as can be seen in the way incentives are utilised across the value chain of solar PV or batteries. Even though production-linked schemes for solar PV were aimed at building a domestic solar power equipment manufacturing ecosystem and reducing import dependence on renewable energy (MNRE 2021), the supply chain and beneficiaries of the scheme are still heavily dependent on imports. It has been reported by companies that, in some cases, up to 90% of their vendors are entities from China, and there is an overall 70% dependence on China for solar equipment (Jai and Pillay 2024; LiveMint 2024). Similarly, the 2021 Scheme for PLI on Advanced Chemistry Cell (ACC) Battery Storage did not do much to reduce import dependence as the lithium supply chain is controlled by China. The scheme has been relaunched with bids invited from mega battery manufacturing facilities of 10 GWh capacity with a budget of Rs 3,620 crore.

5.4.2.2. Diversified Supply Chain

Besides augmenting the production of minerals within the country, diversification of supply sources abroad is an important element of any domestic government's strategy. This operates at both government-to-government (G2G) and government-to-business (G2B) levels. Since several critical mineral reserves and production are not spread widely all over the world, many countries, including countries like the EU, India and the US, which have energy transition goals, are import-dependent for CMET. The supply chain concentration in select countries poses a risk of disruption and price volatility. To overcome this vulnerability and avert these risks, countries have been actively engaging with resource-rich states other than China. As

discussed in Chapter 3, a number of free trade agreements are now dedicating specific provisions or chapters on critical minerals. The use of newer blocks and partnerships for raw material diplomacy is being promoted as a matter of policy and not in some ad hoc manner.

The other means of diversifying and expanding supply sources from abroad is G2B. Indian government's KABIL consortium has been discussing MOUs abroad and has finally signed an agreement for lithium in Argentina. The overseas acquisition of mines or investment in mines abroad attracts a lot of attention, but their value in strengthening the CMET supply chain is not clear. This is especially important as the overall supply chain of solar PV, as well as electrical storage, is scattered, and unless the processing and component manufacturing is secured, getting access to raw unprocessed minerals will help in solving only a part of the problem. Several mining companies operating in India own mines abroad, but that does not necessarily mean cheaper and assured access for end use in India. Thus, the role of overseas mines in strengthening CMET supply needs more analysis beyond the euphoria.

5.4.2.3. Responsible Sourcing and Resource Development

Although conflict between mining companies and local communities is not new, there is an increasing recognition of the need to balance the rights of local communities with that of the right to develop resources or mineral wealth. Instead of the mining versus communities narrative, there is also a realisation that in case of conflict, mining projects can be shut down, stalled or blocked, causing irreparable loss to revenue, time and reputation. Hence, besides government and regulatory licences, social licence to operate is key to mining. This is even more pertinent to critical minerals as time is extremely precious in the race to find more critical minerals for clean energy. Social Licence to Operate (SLO) as a term became common in the last decade or two and refers to the "degree to which a corporation and its activities meet the expectations of local communities, the wider society, and various constituent groups" (Gunningham et al. 2004:313). This societal approval and acceptance of mining activities can reduce conflict and exposure to social risks (Prno and Slocombe 2012). Despite being in addition to and separate from legal measures, SLO is now being integrated into legal frameworks

and instruments (Heffron et al. 2021). The DMF in India, informed consent, community participation in decision-making, and project benefit streams in other jurisdictions are examples of how laws can help achieve SLO.

5.4.2.4. Circularity

Circularity in mining implies operations with a net positive contribution to the environment and society to promote the responsible use and recovery of metals after entry into markets (ICMM 2024). Besides overarching circular economy roadmaps, circularity in minerals is promoted through a range of product and process instruments, such as life cycle reporting, incentives and enabling framework for reuse and recyclability, waste management laws aligned with recycling and recovery principles, industry-specific material recovery targets or process-oriented regulations for reducing waste generation during extraction processes (Toledano et al. 2023).

The ability of recycling to replace the demand for new minerals is different in different scenarios and calculations, but its potential is widely accepted. In order to ensure circularity in CMET, interventions are needed across the value chain, and therefore not only in the mines and minerals industry, but the end-use industry too. Currently, the regulations are more equipped for the mining sector, with some new efforts and directions in the clean energy sector too, for example, domestic e-waste management rules for solar PV or batteries or recovery targets for certain industries as proposed in the EU CRM Act. Ultimately, the two industries will have to be connected to close the loop and be able to adopt circularity.

India's high import dependence on critical minerals needed for solar PV and electrical storage is clear and illustrated amply in the thesis. Besides augmenting domestic production, import resilience in India is being sought through critical mineral partnerships and blocs, bilateral agreements with mineral-rich countries (government to government) and overseas acquisition and agreements (government to business or business to business) with mining companies in resource-rich countries. However, these measures are important for signalling and are yet to yield any tangible results for the critical minerals or energy industry.

A comparative analysis of India, the EU and the US shows that countries either attempt to increase onshoring or friend-shoring of critical minerals. The thesis in this chapter has formulated four pillars of CMET governance, viz., increased supply, diversified supply chain, responsible sourcing and circularity, in order to contribute to the deployment of solar PV and grid-scale electric storage (Figure 5.5). The last few years have seen considerable action on the first two pillars, as is evident in laws, policies, and announcements by the government as well as non-government agencies. However, the third and fourth pillars have received little or no attention, especially in the Indian context. There is a gap in terms of addressing the environmental and social externalities that may be exacerbated as a push is made for increasing and expediting critical minerals mining. Further, while internationally and in the EU, there is a discussion on the importance of recovery and recycling in reducing demand for new minerals, such approaches are missing in the Indian context, barring a reference to it in the reports by a few think tanks and the NITI Aayog for electrical storage.